Week 01

Case Study: Programming Beyond Coding, Why Every Professional Can Benefit

Riya is a first-year university student pursuing a degree in Economics. She believes learning programming is unnecessary because there are already apps and websites for almost every task. During an orientation session, her professor explains that the existence of many applications does not eliminate the need for programming. Just as existing buildings do not prevent people from constructing or improving new ones, existing software can always be enhanced, customized, or automated to solve new problems.

The professor further explains that programming is not limited to computer science. Whether someone studies mathematics, physics, economics, business, or nutrition, programming helps automate repetitive tasks, analyze data, and build customized solutions that are not available in ready-made software.

When students ask why there are so many programming languages, the professor explains that different languages evolved to solve different kinds of problems. Languages such as FORTRAN focused on scientific computing, COBOL on business applications, C emphasized efficiency, C++ introduced object-oriented programming, and Python made programming easier and more accessible. As technology evolves, programming languages and development tools continue to become simpler and more powerful.

To help students understand what programming really means, the professor conducts a classroom activity. He assigns different hand gestures to represent coffee, tea, water, juice, and cookies. Students quickly learn to interpret combinations of gestures, such as "three cups of coffee" or "two cookies." The professor explains that this activity is similar to programming: a programmer gives precise instructions using symbols that a computer has been taught to understand.

By the end of the session, Riya realizes that programming is not just about writing code, it is a problem-solving skill that can provide an advantage in almost any profession.

Q1. Which statement best captures the professor's reason for learning programming despite the availability of many applications?
A. Existing applications eliminate the need for programmers.
B. Programming enables people to create, improve, and customize solutions beyond existing applications.
C. Programming is useful only for developing mobile apps.
D. Learning programming is compulsory for every profession.

Correct Answer: B
Justification: The professor compares software to buildings—existing solutions do not prevent creating better or more customized ones.

Q2. According to the professor, why have multiple programming languages evolved over time?
A. Each country develops its own programming language.
B. Different languages were designed to solve different types of problems and improve

programming.
C. Older programming languages cannot run on modern computers.
D. Every programmer must create a new programming language.

Correct Answer: B
Justification: Programming languages evolved to address different domains and make programming more effective and easier.

Q3. What was the primary purpose of the hand-gesture classroom activity?
A. To improve students' memory.
B. To demonstrate that programming involves giving clear instructions using agreed-upon symbols.
C. To teach sign language.
D. To measure students' reaction speed.

Correct Answer: B
Justification: The activity illustrated that programming is fundamentally about communicating precise instructions through symbols.

Q4. According to the professor, what is the most effective way to become proficient in programming?
A. Memorize every programming language before writing code.
B. Begin with advanced programming problems.
C. Practice regularly, stay motivated, and learn gradually through small steps.
D. Depend entirely on modern programming tools and IDEs.

Correct Answer: C
Justification: The professor emphasizes that programming is a skill developed through consistent practice and gradual learning.

Q5. Which conclusion would Riya most likely draw after attending the session?
A. Programming is a problem-solving skill that can benefit people across many disciplines.
B. Programming is valuable only for software engineers.
C. Programming is primarily about memorizing the syntax of different languages.
D. Existing software packages make programming unnecessary.

Correct Answer: A
Justification: The session highlights that programming is a transferable problem-solving skill applicable to diverse fields such as economics, science, business, and healthcare.

Case Study: Teaching a Robot Efficiently – The Power of Loops

A robotics club is programming a small robot to train for a fitness challenge. The robot must move forward and then return to its starting position. On every round, it should travel a little farther than before. Initially, one student writes separate instructions for every movement: move forward, pause, move backward, pause, and repeats this pattern many times with different distances.

Another student notices that the instructions follow the same pattern repeatedly. Instead of writing dozens of similar commands, she suggests using a loop along with a variable that

stores the current distance. After every round, the variable is increased by a fixed amount, and the loop repeats until all rounds are completed.

The instructor points out that this shorter program is easier to understand, modify, and reuse. Later, instead of fixing the starting distance in the program, they introduce another variable so that the same program can be used for different fitness challenges without changing the main logic.

The students realize that loops are not just about reducing the number of lines of code—they make programs flexible, easier to maintain, and less prone to errors.

Q1. Why did the second student recommend using a loop instead of writing every movement instruction separately?
A. Loops allow repeated actions to be expressed with fewer instructions while preserving the same behavior.
B. Loops make the robot move faster.
C. Loops reduce the distance travelled by the robot.
D. Loops automatically correct programming errors.

Correct Answer: A
Justification: A loop eliminates repetitive code by executing the same sequence multiple times.

Q2. In the improved program, what is the primary purpose of using a variable for the travel distance?
A. To store the robot's battery level.
B. To allow the distance to change automatically after each repetition.
C. To count how many students are programming.
D. To increase the robot's speed.

Correct Answer: B
Justification: The variable stores the current distance and is updated after each iteration, allowing the robot to travel progressively farther.

Q3. The instructor later replaces the fixed starting distance with another variable. What is the main advantage of this change?
A. The robot no longer requires a loop.
B. The same program can easily be reused for different starting distances without rewriting the logic.
C. The program will execute only once.
D. Variables automatically make the program run faster.

Correct Answer: B
Justification: Replacing fixed values with variables increases flexibility and reusability.

Q4. Suppose a student removes the statement that updates the distance variable inside the loop. Which outcome is most likely?
A. The robot will continue repeating the same movement distance in every round.
B. The robot will stop after the first movement.
C. The robot will automatically choose new distances.

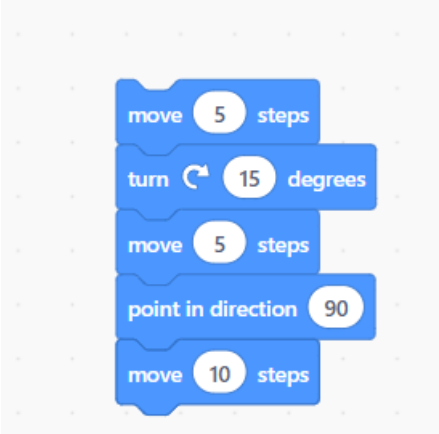
D. The loop will immediately terminate.

Correct Answer: A
Justification: Without updating the variable, its value remains unchanged, so every iteration performs the same movement.

Q5. Which statement best summarizes the role of loops in programming?
A. Loops are useful only for mathematical calculations.
B. Loops replace the need for variables.
C. Loops help automate repetitive tasks and make programs easier to modify and maintain.
D. Loops should be avoided because they make programs difficult to understand.

Correct Answer: C
Justification: Loops improve readability, reduce duplication, and make programs easier to update and extend.

Case Study: Understanding a Scratch Program
A student is learning Scratch and creates the program shown in the figure below. The sprite starts from its initial position and executes each block from top to bottom exactly once. The student expects the sprite to follow the path specified by the movement and direction blocks. Study the Scratch program carefully that is presented below and answer the following questions based on the program shown.



Q1. Which Scratch block determines the direction of the sprite before its final movement of 10 steps?
A. move 5 steps (first block)
B. turn 15 degrees
C. move 5 steps (third block)
D. point in direction 90°

Correct Answer: D
Justification: The point in direction 90° block sets the sprite's direction immediately before the final move 10 steps block.

Q2. What is the purpose of the turn 15 degrees block in this program?
A. It changes the sprite's direction before the second movement.
B. It changes the sprite's position by 15 steps.

C. It changes the direction for the entire program, including after the point in direction block.
D. It makes the sprite move automatically.

Correct Answer: A
Justification: The turn block only changes the current direction. The next movement follows this new direction until another direction block changes it.

Q3. Suppose the point in direction 90° block is removed while keeping all other blocks unchanged. Which statement is most likely to be true?
A. The final movement will continue in the direction established by the earlier turn.
B. The sprite will not execute the last move.
C. The sprite will return to its starting position.
D. The program will stop with an error.

Correct Answer: A
Justification: Without the point in direction block, the sprite keeps its current direction after the 15° turn.

Q4. Which Scratch blocks in the program only change the sprite's position and never its direction?
A. the two move 5 blocks
B. all three move blocks
C. turn and point
D. move 10 and point

Correct Answer: B
Justification: The move blocks change only the sprite's position. They do not alter its facing direction.

Q5. A student accidentally swaps the positions of the turn 15 degrees block and the point in direction 90° block. What is the most likely effect?
A. The sprite will follow exactly the same path.
B. The final movement will occur after turning 15° from 90°, producing a different path.
C. The sprite will not execute the second movement.
D. The program will become an infinite loop.

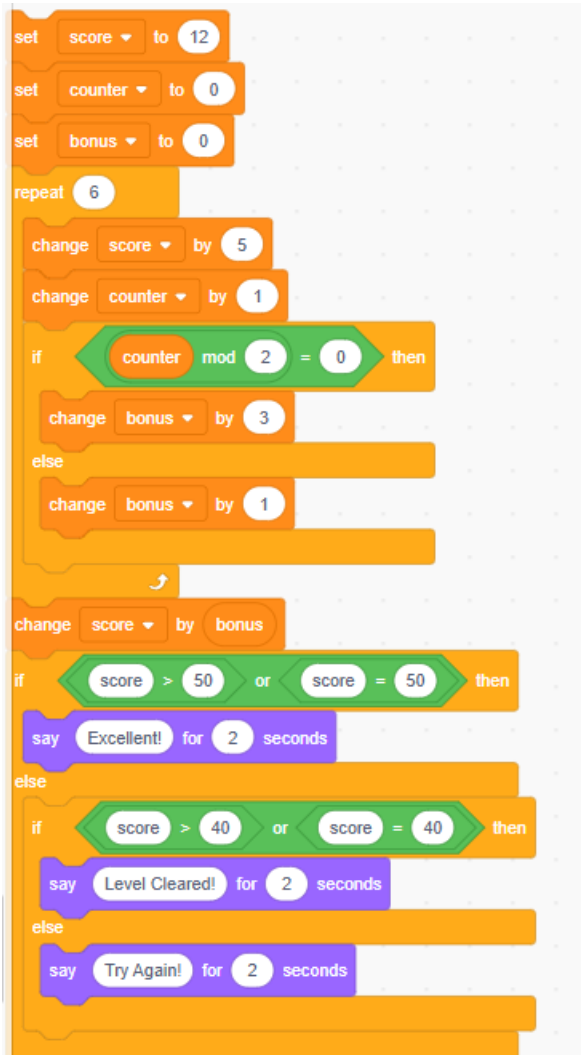
Correct Answer: B
Justification: Scratch executes blocks sequentially. Swapping the two direction blocks changes the direction before the final movement, resulting in a different path.

Case Study: Evaluating a Scratch-Based Game Scoring System

A student is developing a simple game in Scratch to reward players based on their performance. The Scratch program shown in the figure keeps track of three variables: score, counter, and bonus.

When the green flag is clicked, the variables are initialized. The program then repeats a set of instructions several times. During each repetition, the player's score increases, the round counter is updated, and a bonus is awarded depending on whether the current round number is even or odd. After all repetitions are complete, the accumulated bonus is added to the score.

Finally, the program displays one of three messages—"Excellent!", "Level Cleared!", or "Try Again!"—depending on the final score.
Study the Scratch program carefully and answer the following questions.



Q1. Why does the program use the expression counter mod 2 = 0?
A. To check whether the current round number is even.
B. To stop the repeat loop after two iterations.
C. To calculate the total score.
D. To reset the bonus variable.

Correct Answer: A
Justification: The modulo operation returns the remainder after division by 2. A remainder of 0 indicates an even number.

Q2. Which variable is updated in every iteration of the repeat loop, regardless of the outcome of the if-else condition?
A. bonus
B. counter
C. score
D. Both score and counter

Correct Answer: D
Justification: Every iteration increases both score and counter. The bonus changes in both branches, but by different amounts.

Q3. What is the purpose of the statement change score by bonus placed after the repeat loop?
A. It resets the score before displaying the final message.
B. It adds the total bonus accumulated during all iterations to the final score.
C. It increases the bonus by the current score.
D. It determines how many times the loop executes.

Correct Answer: B
Justification: The bonus is accumulated during the loop and added to the score only once after the loop finishes.

Q4. Suppose the statement change score by bonus is moved inside the repeat loop, immediately before the if condition. What would be the most likely effect?
A. The final score would become much larger because the bonus would be added repeatedly.
B. The final score would remain unchanged.
C. The bonus variable would always become zero.
D. The repeat loop would execute fewer times.

Correct Answer: A
Justification: Adding the accumulated bonus during every iteration would repeatedly increase the score, resulting in a much larger final value.

Q5. Which statement best describes how the Scratch program decides which message to display?
A. The message depends only on the value of counter.
B. The message depends only on the value of bonus.
C. The program first computes the final score and then checks the score against the conditions in order.
D. All three messages are displayed one after another.

Correct Answer: C
Justification: After the bonus is added to the score, the program evaluates the conditions sequentially. Once a condition is true, the corresponding message is displayed.

Week 02

Scenario: Student Registration for a Coding Bootcamp

A school is organizing a one-day coding bootcamp for beginners. When students arrive, they register at a computer kiosk. The kiosk asks each student to enter their name and age. It stores this information so it can display a personalized welcome message and generate a participation slip. During registration, if a student notices that they entered incorrect information, they can update it before the final slip is printed. The computer always uses the most recently entered information when displaying the student's details.

Question 1

Why does the registration kiosk ask each student to enter their name?

A. To personalize the welcome message and participation slip.

B. To make the computer process information faster.

C. To install the coding software.

D. To increase the computer's storage.

Answer: A

Justification: The student's name is collected so the computer can use it in personalized messages and printed information.

Question 2

A student first enters their age as **15**, but later realizes it should be **16** and updates it before the participation slip is printed. Which age will appear on the slip?

A. 15

B. Both 15 and 16

C. 16

D. No age will be displayed.

Answer: C

Justification: The latest value replaces the previous one, so the updated age is used.

Question 3

The computer displays a greeting that includes the student's name without asking for it again. How is this possible?

A. The computer remembers every student's name forever.

B. The name is stored and reused whenever needed during the program.

C. The greeting works only for the first student.

D. The student secretly enters the name again.

Answer: B

Justification: The program stores the entered name and can use it multiple times without asking again.

Question 4

A student accidentally enters the name **Riya** instead of **Diya**. Before the participation slip is printed, the mistake is corrected. What is the most likely outcome?

A. The slip will show Riya.

B. The slip will show both Riya and Diya.

C. The slip will show Diya.

D. The slip will not display any name.

Answer: C

Justification: Once the information is updated, the program uses the corrected value instead of the old one.

Question 5

Why is it useful for the program to store information like a student's name and age?

A. It allows the same information to be used multiple times without asking the student again.

B. It permanently stores the information on every computer.

C. It prevents the program from displaying messages.

D. It eliminates the need for user input in all future programs.

Answer: A

Justification: Storing information allows the program to reuse it whenever needed during its execution, making the interaction more efficient.

Scenario: Ordering Food at a School Cafeteria

A school cafeteria has introduced a self-service ordering kiosk. Students enter their name and the food item they would like to order. The kiosk stores this information and displays a

confirmation message. Before the order is finalized, one student changes their order from **Sandwich** to **Pasta**. The kiosk updates the stored value and prints the final order receipt. The Python program running on the kiosk is shown below.

```
1. print("Welcome to the School Cafeteria!")
2. customer = input("Enter your name: ")
3. order = input("Enter your food item: ")
4. print("Hello", customer)
5. print("You ordered", order)
6. order = "Pasta"
7. print("Updated Order:", order)
8. print("Thank you for your order,", customer)
```

Question 1

Which line collects the name of the customer?

A. Line 1

B. Line 2

C. Line 3

D. Line 8

Answer: B

Justification: Line 2 uses input() to ask the customer to enter their name.

Question 2

A student initially enters **Sandwich** as the food item. What will be displayed in **Line 7**?

A. Sandwich

B. Sandwich Pasta

C. Pasta

D. No food item will be displayed.

Answer: C

Justification: Line 6 changes the value of order to **Pasta**, so the updated value is printed.

Question 3

Which variable stores the name of the customer throughout the program?

A. order

B. customer

C. input

D. print

Answer: B

Justification: The variable customer stores the user's name and is used whenever the program needs it.

Question 4

If the customer enters **Neha** as the name, what will be displayed in **Line 8**?

A. Thank you for your order, customer

B. Thank you for your order, Neha

C. Thank you for your order, Pasta

D. Thank you for your order

Answer: B

Justification: The variable customer contains the value **Neha**, which is displayed in the final message.

Question 5

Why is the variable order updated in Line 6 instead of creating another variable?

A. A variable can only store one value at a time, and the latest value replaces the previous one.

B. Python does not allow more than one variable in a program.

C. The original value cannot be changed.

D. Updating a variable permanently changes all future programs.

Answer: A

Justification: Assigning a new value to a variable replaces its previous value, so the program always uses the latest information.

Scenario: Online Bookstore Checkout

An online bookstore is developing a simple checkout system. When a customer selects a book, the system asks for the book's price. Since the entered value comes from the keyboard, the system first converts it into a number before performing any calculations. A **10%**

discount is then applied to the original price to calculate the final amount the customer has to pay.

While testing the program, the developers also discover that if they forget to convert the entered value into a number, mathematical calculations do not work as expected. They also learn that Python is very particular about following the correct syntax and indentation rules. Even a small mistake, such as forgetting a colon or incorrect indentation, can prevent the program from running.

Question 1

Why does the program convert the customer's entered price into a number before calculating the discount?

A. To make the input appear on the screen.

B. To allow mathematical operations such as multiplication.

C. To permanently save the customer's purchase history.

D. To reduce the size of the program.

Answer: B

Justification: Keyboard input is initially treated as text. Converting it to a number allows arithmetic operations like calculating discounts.

Question 2

A customer enters the price of a book as **₹800**. After applying a **10% discount**, what amount should the customer pay?

A. ₹720

B. ₹780

C. ₹880

D. ₹700

Answer: A

Justification: A 10% discount means the customer pays 90% of the original price: ₹800 × 0.9 = ₹720.

Question 3

During testing, a developer forgets to convert the entered price into a number before performing calculations. What is the most likely outcome?

A. The calculations may fail or produce unexpected results.

B. The discount becomes 20% automatically.

C. The program permanently deletes the input.

D. Python automatically fixes the mistake.

Answer: A

Justification: Without converting the input into a numeric type, Python treats it as text, so arithmetic operations cannot be performed correctly.

Question 4

Another developer accidentally removes the required colon (:) from a Python statement. What is the most likely result?

A. The program will ignore the missing colon and continue.

B. Python will report a syntax error.

C. The program will apply the wrong discount.

D. The program will ask the user to re-enter the price.

Answer: B

Justification: Python follows strict syntax rules. Missing required symbols, such as a colon, results in a syntax error.

Question 5

Which of the following statements are **correct** about the bookstore's discount program? *(Select all that apply.)*

A. User input must often be converted into a number before performing arithmetic calculations.

B. Python requires correct syntax and indentation for a program to run properly.

C. A 10% discount means the customer pays 90% of the original price.

D. Python automatically treats every keyboard input as a number.

Correct Answers: A, B, C

Justification:

- **A:** Correct. User input is initially treated as text and often needs conversion.
- **B:** Correct. Python is sensitive to syntax and indentation.
- **C:** Correct. Paying after a 10% discount means paying 90% of the original price.
- **D:** Incorrect. Keyboard input is treated as text unless explicitly converted.

Scenario: Morning Exercise Counter

A school has installed a digital exercise board in the playground. Every morning, the board displays the number of jumping jacks students should perform. It starts from **1** and continues until the teacher decides to stop the session. The board repeatedly displays the next exercise number as long as the teacher chooses to continue.

The following Python program is used for the exercise board.

```
1. print("Morning Exercise Begins")
2. count = 1
3. continue_session = 1
4.
5. while continue_session == 1:
6.     print("Do Exercise", count)
7.     count = count + 1
8.     continue_session = int(input("Continue? (1 = Yes, 0 = No): "))
9.
10. print("Exercise Session Over")
```

Question 1

What is displayed **first** when the program starts?

A. Do Exercise 1

B. Continue? (1 = Yes, 0 = No)

C. Morning Exercise Begins

D. Exercise Session Over

Answer: C

Justification: Line 1 executes before the loop begins, displaying the welcome message.

Question 2

The teacher enters **1** three times and then enters **0**. Which exercise numbers will be displayed?

A. 1, 2, 3

B. 1, 2, 3, 4

C. 0, 1, 2, 3

D. 2, 3, 4

Answer: B

Justification: The program displays Exercise **1** initially. Each time the teacher enters **1**, the next exercise number is displayed. After entering **0**, the loop stops, so the displayed numbers are **1, 2, 3, and 4**.

Question 3

Which statements about the program are correct? *(Select all that apply.)*

A. The value of count increases by 1 during each repetition.

B. The while loop repeats only while continue_session is equal to 1.

C. The message "Exercise Session Over" is displayed after the loop ends.

D. The value of count decreases during each repetition.

Correct Answers: A, B, C

Justification:

- **A:** Correct. Line 7 increases count by 1.
- **B:** Correct. The loop condition is checked in Line 5.
- **C:** Correct. Line 10 runs after the loop finishes.
- **D:** Incorrect. count never decreases.

Question 4

Why is a **while loop** used in this program?

A. Because the number of repetitions depends on the teacher's choice.

B. Because the program always repeats exactly 10 times.

C. Because Python cannot print numbers without a while loop.

D. Because variables cannot be used inside a for loop.

Answer: A

Justification: The teacher decides when to stop the exercise session, so the number of repetitions is not fixed beforehand.

Question 5

Which of the following statements correctly compare **for loops** and **while loops**? *(Select all that apply.)*

A. A for loop is useful when the number of repetitions is known in advance.

B. A while loop continues until its condition becomes false.

C. A while loop always executes exactly 10 times.

D. Both for and while loops are used to repeat tasks.

Correct Answers: A, B, D

Justification:

- **A:** Correct. for loops are commonly used for a fixed number of repetitions.
- **B:** Correct. A while loop keeps running as long as its condition is true.
- **C:** Incorrect. A while loop may run any number of times depending on the condition.
- **D:** Correct. Both loop types are repetition constructs used in programming.

Week 03

Scenario: Managing the School Library

A school library maintains a digital catalog of all its books using a Python list. As new books arrive, the librarian updates the catalog by adding new titles or inserting important reference books at specific positions. Sometimes multiple copies of the same book are available, so the librarian checks how many times a title appears in the list. To help students find books more easily, the catalog is sorted alphabetically. During special exhibitions, the sorted list is displayed in reverse alphabetical order. Occasionally, the librarian also displays only a selected range of books, such as books belonging to a particular shelf or category, instead of showing the entire catalog.

Answer the following questions based on the scenario.

Q1. Why does the librarian sort the catalog of books?

A. To remove duplicate book titles

B. To make the books easier to organize and locate

C. To increase the number of books in the catalog

D. To replace old books with new ones

Answer: B

Justification: Sorting arranges the book titles in a meaningful order, making them easier to organize, search, and analyze.

Q2. After sorting the catalog alphabetically, the librarian wants to display the books from Z to A. Which operation is most suitable?

A. Count

B. Append

C. Reverse

D. Length

Answer: C

Justification: Reversing a sorted list changes the order from ascending (A–Z) to descending (Z–A).

Q3. The librarian wants to know how many copies of the book *Python Basics* are present in the catalog. Which list operation is most appropriate?

A. Length

B. Reverse

C. Insert

D. Count

Answer: D

Justification: The count() operation returns the number of times a particular item appears in the list.

Q4. A newly purchased encyclopedia must appear as the second item in the catalog instead of being added at the end. Which operation should be used?

A. Insert

B. Append

C. Sort

D. Reverse

Answer: A

Justification: The insert() operation places an item at a specified position in the list.

Q5. The librarian wants to know the total number of books currently stored in the catalog. Which operation provides this information?

A. Count

B. Length

C. Reverse

D. Slice

Answer: B

Justification: The length operation returns the total number of elements present in the list.

Scenario: Managing Participants for a Workshop

A training institute maintains a list of participant names for a Python workshop. As registrations are received, new participants are added to the list. A late VIP registration is inserted at a specific position. Before printing attendance sheets, the participant names are arranged alphabetically. Finally, only a selected group of participants is displayed for one workshop batch.

Study the following program and answer the questions.

```
1. participants = ["Riya", "Aman", "Kiran", "Neha", "Aman"]
2. participants.append("Sana")
3. participants.insert(1, "Bhavna")
4. participants.sort()
5. batch = participants[1:5]
6. print(participants)
7. print(batch)
```

Q1. What will be the output of line 6?

A. ['Aman', 'Aman', 'Bhavna', 'Kiran', 'Neha', 'Riya', 'Sana']

B. ['Riya', 'Bhavna', 'Aman', 'Kiran', 'Neha', 'Aman', 'Sana']

C. ['Aman', 'Bhavna', 'Kiran', 'Neha', 'Riya', 'Sana']

D. ['Bhavna', 'Aman', 'Aman', 'Kiran', 'Neha', 'Riya', 'Sana']

Answer: A

Justification: After appending and inserting, the list is sorted alphabetically before being printed.

Q2. What is stored in the variable batch after line 5?

A. ['Aman', 'Bhavna', 'Kiran', 'Neha']

B. ['Bhavna', 'Kiran', 'Neha', 'Riya']

C. ['Aman', 'Aman', 'Bhavna', 'Kiran']

D. ['Kiran', 'Neha', 'Riya', 'Sana']

Answer: A

Justification: Slicing participants[1:5] returns elements at indices 1, 2, 3, and 4.

Q3. (MSQ) Which of the following statements are true after line 4?

A. The participant Sana has been added to the list.

B. The name Aman appears twice.

C. The list contains 6 participants.

D. The names are arranged alphabetically.

Answer: A, B, D

Justification:

- A: True. append() adds Sana.
- B: True. Aman appears twice.
- C: False. The list contains 7 participants.
- D: True. sort() arranges the names alphabetically.

Q4. Which line inserts the participant Bhavna at a specific position in the list?

A. Line 1

B. Line 2

C. Line 3

D. Line 5

Answer: C

Justification: insert(1, "Bhavna") places Bhavna at index 1 before sorting.

Q5. Which operation is demonstrated by line 5?

A. Appending a new participant

B. Sorting the participant list

C. Counting duplicate names

D. Extracting a portion of the participant list

Answer: D

Justification: The slicing operation creates a new list containing only a selected range of participants.

Scenario: Traffic Signal Announcement System

A city traffic department is developing a digital announcement system for a pedestrian crossing. The system displays numbers from 1 to 30. To make the display more informative, it follows these rules:

- If the number is divisible by 3, it displays "Walk" instead of the number.
- If the number is divisible by 5, it displays "Wait" instead of the number.
- If the number is divisible by both 3 and 5, it displays "WalkWait".
- Otherwise, it displays the number itself.

A programmer uses a loop and conditional statements to implement this logic.

Answer the following questions based on the scenario.

Q1. What should the system display when the number 15 is reached?

A. 15

B. Walk

C. Wait

D. WalkWait

Answer: D

Justification: Since 15 is divisible by both 3 and 5, the system displays WalkWait.

Q2. Which of the following numbers will display "Walk"?

A. 8

B. 12

C. 20

D. 25

Answer: B

Justification: 12 is divisible by 3 but not by 5.

Q3. A programmer first checks whether a number is divisible by 3, then checks whether it is divisible by 5, and only after that checks whether it is divisible by both 3 and 5. What problem is likely to occur?

A. Numbers divisible by both 3 and 5 will be displayed as Walk instead of WalkWait.

B. Numbers divisible by 5 will never be displayed.

C. The loop will never stop.

D. The program will print every number twice.

Answer: A

Justification: The first condition is satisfied for numbers like 15, so the later check for both conditions is never reached.

Q4. Which programming concept allows the system to process every number from 1 to 30 automatically?

A. Function

B. List

C. Loop

D. Variable

Answer: C

Justification: A loop repeats the same set of instructions for each number in the specified range.

Q5. Which statement best explains why the condition "divisible by both 3 and 5" should be checked before checking divisibility by 3 or 5 individually?
A. It reduces the number of iterations in the loop.
B. It ensures that numbers satisfying both conditions receive the correct output.
C. It changes the range of numbers processed.
D. It makes the program run without using loops.
Answer: B
Justification: Checking the combined condition first prevents numbers like 15 from being incorrectly classified as only divisible by 3 or only divisible by 5.

Scenario: Choosing the Best Estimate for a Wildlife Survey

A wildlife research team wants to estimate the number of birds nesting in a protected forest. Since counting every bird is difficult, they ask 120 volunteers to independently estimate the number of nests. The estimates vary widely because some volunteers underestimate while others overestimate the actual value. The researchers analyze the estimates by sorting them, removing extreme values, and computing different aggregate measures. They also visualize the estimates using Matplotlib to better understand the spread of the data.
Answer the following questions based on the scenario.

Q1. (MCQ)
Why did the researchers ask a large number of volunteers instead of only one wildlife expert?
A. To reduce the size of the dataset
B. Because a larger group can produce a reliable collective estimate
C. Because volunteers always know the exact answer
D. To avoid using statistical measures
Answer: B
Justification: The wisdom of crowds suggests that combining many independent estimates often leads to a reliable result.

Q2. (MCQ)

Which statistical measure is specifically designed to reduce the effect of extremely high and extremely low estimates?

A. Maximum

B. Median

C. 10% Trimmed Mean

D. Range

Answer: C

Justification: A trimmed mean removes a percentage of the smallest and largest values before calculating the average.

Q3. (MCQ)

The researchers create a histogram of all the estimates using Matplotlib. What is the main purpose of this graph?

A. To sort the estimates automatically

B. To visualize how the estimates are distributed

C. To calculate the trimmed mean

D. To remove incorrect estimates

Answer: B

Justification: A histogram helps visualize the distribution and frequency of estimates.

Q4. (MSQ)

Which of the following statements about 10% trimmed mean are correct?

A. The data should be sorted before removing extreme values.

B. The smallest and largest 10% of estimates are removed.

C. Every estimate is included in the final calculation.

D. It helps reduce the influence of outliers.

Answer: A, B, D

Justification:

- A: Correct. Sorting is required to identify the extreme values.
- B: Correct. Both the lowest and highest 10% are removed.
- C: Incorrect. Extreme values are excluded.
- D: Correct. This reduces the effect of unusually high or low estimates.

Q5. (MSQ)

The researchers use the following statement before calculating the trimmed mean:

estimates.sort()

Which of the following are true about this statement?

A. It arranges the estimates in ascending order.

B. It changes the original list of estimates.

C. It creates a new sorted copy while leaving the original unchanged.

D. It helps identify the smallest and largest estimates for trimming.

Answer: A, B, D

Justification:

- A: Correct. sort() arranges elements in ascending order.
- B: Correct. sort() modifies the original list.
- C: Incorrect. sort() does not create a new list; sorted() does.
- D: Correct. Once sorted, extreme values are easy to locate and remove.

Week 04

Scenario: Birthday Collision Simulator

A data science club is developing a Python program to demonstrate the Birthday Paradox, which shows that in a surprisingly small group of people, two individuals are likely to share the same birthday. Since collecting real birthday data is impractical, the program randomly generates birthdays for 50 people. It first generates a random year, month, and day while ensuring that the generated date is valid by checking whether the selected year is a leap year. The birthdays are stored as datetime.date objects and then converted into their corresponding day of the year (1–365 or 366), since only the day and month are relevant for detecting birthday matches. These values are stored in a list, sorted in ascending order, and scanned for repeated values. Each repeated value represents a birthday collision. Because the birthdays are generated randomly, running the program multiple times produces different numbers of collisions, helping students understand the Birthday Paradox through simulation.

MCQ 1

Why does the program check whether a randomly generated year is a leap year?

A. To ensure that February is assigned the correct number of days.

B. To reduce the number of duplicate birthdays.

C. To generate birthdays in chronological order.

D. To calculate the person's age.

Answer: A

Justification: Leap years determine whether February has 28 or 29 days.

MCQ 2

Which Python library is primarily responsible for creating valid calendar dates in the program?

A. random

B. math

C. datetime

D. calendar

Answer: C

Justification: The datetime library is used to represent birthdays as valid date objects.

MCQ 3

Why does the program convert each birthday into its "day of the year" value?

A. To calculate the age of each person.

B. To compare birthdays without considering the birth year.

C. To reduce the amount of memory used.

D. To make sorting unnecessary.

Answer: B

Justification: Only the day and month matter when checking for birthday collisions.

MCQ 4

Why is the list of day-of-year values sorted before checking for collisions?

A. To generate birthdays more efficiently.

B. To remove invalid dates automatically.

C. To arrange birthdays by year.

D. To make duplicate values easier to identify.

Answer: D

Justification: Sorting places identical values next to each other, making collisions easy to detect.

MCQ 5

Why does the program usually produce a different number of collisions each time it is executed?

A. The random birthday generation creates a different set of birthdays in every run.

B. The leap year calculation changes during execution.

C. The sorting algorithm produces a different order every time.

D. The datetime library stores dates differently in each execution.

Answer: A

Justification: Different randomly generated birthdays result in different collision counts.

Scenario: Digital Dobble Card Generator

A game development team is creating a digital version of the Dobble card game using Python. Each card contains five symbols, represented by letters of the alphabet. The program randomly generates two cards such that exactly one symbol is common between them, while

all other symbols are unique. To achieve this, it uses Python's string library to obtain a pool of symbols and the random library to select symbols and their positions. Once a symbol is placed on a card, it is removed from the available pool to prevent duplicates. After generating the cards, the program displays them and asks the player to identify the common symbol. If the player's input matches the common symbol, the program prints "Correct"; otherwise, it prints "Wrong". During testing, the developers also discovered that generating random indices from 0 to 5 for a five-element list caused an IndexError, which was fixed by generating indices from 0 to 4.

MCQ 1

What is the primary objective of the Python program?

A. To generate two cards with exactly one common symbol.

B. To sort symbols alphabetically before displaying them.

C. To count the number of vowels on each card.

D. To generate identical cards every time.

Answer: A

Justification: The program is designed so that the two cards share exactly one common symbol.

MCQ 2 (MSQ)

Which of the following steps help ensure that the two cards contain exactly one common symbol? *(Select all that apply.)*

A. Removing the common symbol from the symbol pool after placing it.

B. Removing every newly assigned symbol from the available symbol pool.

C. Randomly choosing only one symbol to be shared between both cards.

D. Sorting the cards after they are generated.

Answer: A, B, C

Justification: Removing used symbols prevents extra matches, while random positions determine where the single common symbol appears. Sorting does not affect the card contents.

MCQ 3

Why does the program use random.choice()?

A. To sort the available symbols.

B. To select a symbol randomly from the available symbol list.

C. To compare the two cards.

D. To convert uppercase letters into lowercase letters.

Answer: B

Justification: random.choice() selects a random element from a list.

MCQ 4

Why was random.randint(0, 5) changed to random.randint(0, 4) while selecting card positions?

A. The cards contain only four symbols.

B. The random library cannot generate the value 5.

C. A five-element list has valid indices only from 0 to 4.

D. Python lists always begin with index 1.

Answer: C

Justification: A list with five elements has indices 0 through 4; index 5 is out of range.

MCQ 5

What does the program do after displaying both generated cards?

A. It automatically highlights the common symbol.

B. It asks the player to identify the common symbol.

C. It generates another pair of cards immediately.

D. It removes all symbols from both cards.

Answer: B

Justification: The player is prompted to enter the common symbol, and the program checks whether the answer is correct.

Scenario: Building the First Prototype of a Magic Square Generator

A software development team is creating a Python program to generate odd-sized Magic Squares. Rather than implementing the complete algorithm immediately, they decide to build and test the program incrementally. In the first prototype, the program accepts the size of the square (n), creates an n × n matrix initialized with zeros, determines the starting position for the first number using the Magic Square algorithm, places the number 1 at that position, and then displays the matrix for verification. Once this initial step works correctly, the team plans

to extend the program to place the remaining numbers by following the movement rules of the algorithm.

```
n = 3

magic_square = [[0 for j in range(n)] for i in range(n)]

row = n // 2
col = n - 1

magic_square[row][col] = 1

for r in magic_square:
    print(r)
```

MCQ 1

Which programming construct is primarily responsible for creating the two-dimensional matrix in the prototype?

A. Nested list comprehension

B. Nested if-else statements

C. Nested while loops

D. Function recursion

Answer: A

Justification: The matrix is created using a nested list comprehension that generates rows and columns.

MCQ 2

If the value of n is changed from 3 to 5, which statement remains true without modifying the rest of the program?

A. The number 1 will always be placed in the center of the matrix.

B. The number 1 will always be placed in the middle row of the last column.

C. The matrix will automatically become a valid Magic Square.

D. The program will generate numbers from 1 to 25 automatically.

Answer: B

Justification: The starting position is determined by the algorithm and remains the middle row of the last column for any odd-sized matrix.

MCQ 3

Why is the matrix initialized with zeros instead of leaving it empty?

A. To allow the program to identify cells that have not yet been assigned a value.

B. To ensure that every row initially has the same sum.

C. To improve the speed of printing the matrix.

D. To prevent duplicate numbers from being generated.

Answer: A

Justification: Zero acts as a placeholder, making it easy to distinguish occupied and unoccupied cells.

MCQ 4

What is the primary limitation of this prototype compared to the complete Magic Square algorithm?

A. It cannot create matrices larger than 3 × 3.

B. It places only the first number and does not determine positions for the remaining numbers.

C. It cannot display the matrix.

D. It requires recursion to execute.

Answer: B

Justification: The prototype demonstrates only the initialization and placement of the first value.

MSQ 5

Which of the following statements correctly describe the prototype program? *(Select all that apply.)*

A. It allocates memory for an n × n matrix before placing any numbers.

B. It determines the initial position of the first number using the Magic Square algorithm.

C. It uses nested iteration while creating the matrix.

D. It verifies that every row, column, and diagonal has the same sum.

Answer: A, B, C

Justification: The prototype creates the matrix, determines the starting position, and uses nested iteration for initialization, but it does not validate or construct the complete Magic Square.

Scenario: Movie Guessing Game

A movie streaming platform is developing a Movie Guessing Game to improve user engagement. The game randomly selects a movie title from a predefined list and displays a masked version of it, where every alphabetic character is replaced with * while spaces remain visible. The player repeatedly guesses one letter at a time. If the guessed letter is present in the movie title, all its occurrences are revealed. Otherwise, the player is informed that the letter is not present. The game continues until the player successfully reveals the complete movie title.

```
import random

movies = [
    "THE LION KING",
    "3 IDIOTS",
    "DANGAL",
    "LAGAAN",
    "INTERSTELLAR"
]


def create_question(movie):
    question = ""

    for ch in movie:
        if ch == " ":
            question += " "
        elif ch.isalpha():
            question += "*"
        else:
            question += ch

    return question
```

```
def is_present(letter, movie):
    return movie.count(letter) > 0


def unlock(question, movie, letter):
    result = ""

    for i in range(len(movie)):
        if movie[i] == letter:
            result += letter
        else:
            result += question[i]

    return result


movie = random.choice(movies)

question = create_question(movie)
```

```
print("Guess the movie:")
print(question)

while question != movie:

    letter = input("Enter a letter: ").upper()

    if is_present(letter, movie):
        question = unlock(question, movie, letter)
        print("Correct!")
    else:
        print("Letter not found.")

    print(question)

print("Congratulations! You guessed the movie.")
```

MCQ 1

Which function is responsible for creating the initial masked movie title displayed to the player?

A. unlock()

B. is_present()

C. create_question()

D. random.choice()

Answer: C

Justification: create_question() replaces alphabetic characters with * while preserving spaces and other characters.

MCQ 2

The player enters a letter that is not present in the selected movie title. What happens next?

A. The movie title is selected again.

B. The current masked title remains unchanged, and a message is displayed.

C. The game terminates immediately.

D. All hidden letters are revealed.

Answer: B

Justification: If the guessed letter is absent, the program displays "Letter not found." and continues with the existing masked title.

MCQ 3

Why does the unlock() function receive both the original movie title and the current masked title as parameters?

A. To compare their lengths before displaying the result.

B. To generate a new random movie title.

C. To reveal only the newly guessed letters while preserving previously revealed letters.

D. To convert the movie title into uppercase.

Answer: C

Justification: The original title identifies matching letters, while the masked title preserves letters revealed in earlier guesses.

MCQ 4

Which condition causes the game loop to terminate?

A. The player enters an incorrect letter.

B. All letters in the movie title have been revealed.

C. The player guesses five incorrect letters.

D. The movie list becomes empty.

Answer: B

Justification: The loop continues until the masked title becomes identical to the original movie title.

MSQ 5

Which of the following statements about the program are correct? *(Select all that apply.)*

A. random.choice() is used to select one movie title from the list.

B. isalpha() ensures that spaces and non-alphabetic characters are replaced with *.

C. The unlock() function may reveal multiple occurrences of the guessed letter in a single attempt.

D. The player's input is converted to uppercase before it is checked against the movie title.

Answer: A, C, D

Justification: The program randomly selects a movie, converts the player's input to uppercase, and reveals every occurrence of a correctly guessed letter. isalpha() masks only alphabetic characters, leaving spaces and other non-alphabetic characters unchanged rather than replacing them.

Week 05 – Assignments

Scenario: Community Library Book Tracker

A community library keeps track of the number of books borrowed from different categories. At the end of the day, the librarian wants to update the records based on newly borrowed books and determine which category had the highest number of borrowings.

Study the following Python program carefully and answer the questions that follow.

```
1.  borrow_count = {
2.      "Fiction": 12,
3.      "Science": 8,
4.      "History": 5,
5.      "Technology": 10
6.  }

7.  updates = {
8.      "Science": 4,
9.      "History": 2,
10.     "Art": 6
11. }

12. for category, count in updates.items():
13.     if category in borrow_count:
14.         borrow_count[category] += count
15.     else:
16.         borrow_count[category] = count

17. highest = ""
18. max_books = -1

19. for category, count in borrow_count.items():
20.     if count > max_books:
21.         max_books = count
22.         highest = category

23. print(borrow_count)
24. print(highest, max_books)
```

Q1. (MCQ)

How many key-value pairs are present in borrow_count immediately after the completion of Line 16?

A. 4

B. 5

C. 6

D. 7

Answer: B

Justification:

Initially, borrow_count contains four key-value pairs. During the update process, "Science" and "History" are updated, while "Art" is added as a new key. Therefore, the dictionary contains five key-value pairs after Line 16.

Q2. (MCQ)

Which statement correctly describes the execution of the else block (Lines 15–16)?

A. It executes for every category in updates.

B. It executes only when a category already exists in borrow_count.

C. It executes only when a category is not present in borrow_count.

D. It never executes because all categories already exist.

Answer: C

Justification:

The else block is executed only when the condition category in borrow_count evaluates to False. In this program, only the category "Art" is absent from borrow_count, so the else block executes exactly once.

Q3. (MCQ)

Immediately after executing Line 21 for the first time, what is the value of max_books?

A. -1

B. 10

C. 12

D. 7

Answer: C

Justification:

Before the second loop begins, max_books is initialized to -1. The first category processed is "Fiction" with a borrowing count of 12. Since 12 > -1, max_books is updated to 12.

Q4. (MCQ)

Which variable stores the largest borrowing count after the program completes execution?

A. highest

B. category

C. count

D. max_books

Answer: D

Justification:

The variable max_books stores the highest borrowing count encountered during the traversal of the dictionary. The variable highest stores the corresponding category name, while category and count represent the current key and value during iteration.

Q5. (MSQ)

Which of the following statements are correct?

A. The values of "Science" and "History" are increased.

B. The value of "Technology" remains unchanged.

C. Exactly one new key is added to the dictionary.

D. The final dictionary contains fewer keys than the initial dictionary.

Answer: A, B, C

Justification:

- A: Correct. "Science" is updated from 8 to 12, and "History" is updated from 5 to 7.
- B: Correct. "Technology" is not present in the updates dictionary, so its value remains unchanged.
- C: Correct. "Art" is the only new category added to borrow_count.
- D: Incorrect. The dictionary grows from four keys to five keys after the addition of "Art".

Scenario: Rock–Paper–Scissors Championship

A coding club is organizing a Rock–Paper–Scissors championship. The choices made by both players are stored in two lists. A dictionary is used to define which move defeats another. The program compares the choices in each round, updates the scores, and declares the overall winner.

Study the following program and answer the questions.

```
wins_against = {
    "Rock": "Scissors",
    "Paper": "Rock",
    "Scissors": "Paper"
}

playerA = ["Rock", "Paper", "Scissors", "Rock", "Paper"]
playerB = ["Scissors", "Rock", "Paper", "Rock", "Scissors"]

score = {
    "Player A": 0,
    "Player B": 0
}

for i in range(len(playerA)):
    if playerA[i] == playerB[i]:
        continue
    elif wins_against[playerA[i]] == playerB[i]:
        score["Player A"] += 1
    else:
        score["Player B"] += 1

print(score)

if score["Player A"] > score["Player B"]:
    print("Player A Wins")
elif score["Player B"] > score["Player A"]:
    print("Player B Wins")
else:
    print("Match Draw")
```

Q1. (MCQ)

What is the purpose of the dictionary wins_against?

A. It stores the scores of both players.

B. It stores the choices made by each player.

C. It stores which move defeats another move.

D. It stores the winner of each round.

Answer: C

Justification:

The dictionary maps each move to the move it defeats. For example, "Rock" maps to "Scissors" because Rock beats Scissors.

Q2. (MCQ)

What is the value of the score dictionary after the loop finishes?

A. {"Player A": 2, "Player B": 2}

B. {"Player A": 3, "Player B": 1}

C. {"Player A": 4, "Player B": 0}

D. {"Player A": 1, "Player B": 3}

Answer: B

Justification:

Player A wins Rounds 1, 2, and 3, while Player B wins Round 5. Round 4 is a draw, so the final score is {"Player A": 3, "Player B": 1}.

Q3. (MCQ)

What does the expression wins_against[playerA[i]] return?

A. The current score of Player A.

B. The move that defeats Player A's current move.

C. The move defeated by Player A's current move.

D. The current round number.

Answer: C

Justification:

The expression retrieves the value associated with Player A's current move from the dictionary. For example, if Player A chooses "Paper", it returns "Rock".

Q4. (MCQ)

Which statement is printed at the end of the program?

A. Player A Wins

B. Player B Wins

C. Match Draw

D. No output is printed.

Answer: A

Justification:

Player A scores 3 points, while Player B scores 1 point. Since Player A has the higher score, the program prints "Player A Wins".

Q5. (MSQ)

Which of the following statements are correct?

A. The dictionary score is updated during program execution.

B. The dictionary wins_against is modified during the program.

C. The program records one drawn round.

D. The statement score["Player A"] += 1 updates the value associated with the key "Player A".

Answer: A, C, D

Justification:

The score dictionary is updated whenever a player wins a round, while wins_against is used only for lookups. One round ends in a draw, and score["Player A"] += 1 increments Player A's score by one.

Library Book Search System

A digital library stores book IDs in a sorted list. When a student searches for a book, the librarian first performs a Linear Search to locate the book. Later, the system uses a Binary Search to improve search efficiency on the same sorted list.

Study the following program and answer the questions.

```
1. book_ids = [105, 112, 118, 126, 134, 142, 150, 163]
2. target = 134
3.
4. # Linear Search
5. linear_index = -1
6. for i in range(len(book_ids)):
7.     if book_ids[i] == target:
8.         linear_index = i
9.         break
10.
11. # Binary Search
12. low = 0
13. high = len(book_ids) - 1
14. binary_index = -1
15.
16. while low <= high:
17.     mid = (low + high) // 2
18.
19.     if book_ids[mid] == target:
20.         binary_index = mid
21.         break
22.     elif target < book_ids[mid]:
23.         high = mid - 1
24.     else:
25.         low = mid + 1
26.
27. print("Linear Search:", linear_index)
28. print("Binary Search:", binary_index)
```

Q1. (MCQ)

Why is the list book_ids arranged in ascending order?

A. To reduce memory usage.

B. Because Binary Search requires the data to be sorted.

C. Because Linear Search requires the data to be sorted.

D. To avoid duplicate values.

Answer: B

Justification:

Binary Search works correctly only when the elements are sorted. Linear Search does not require a sorted list.

Q2. (MCQ)

What is the output of the program?

A. Linear Search: 3 and Binary Search: 3

B. Linear Search: 4 and Binary Search: 4

C. Linear Search: 5 and Binary Search: 5

D. Linear Search: -1 and Binary Search: -1

Answer: B

Justification:

The value 134 is present at index 4 in the list. Both search algorithms locate the same element and return the same index.

Q3. (MCQ)

During the first iteration of the Binary Search loop, what is the value of mid?

A. 2

B. 3

C. 4

D. 5

Answer: B

Justification:

Initially, low = 0 and high = 7. Therefore, mid = (0 + 7) // 2 = 3.

Q4. (MCQ)

Suppose the statement break on Line 9 is removed. Which statement is correct?

A. The program produces an error.

B. The Linear Search continues checking the remaining elements even after finding the target.

C. The Binary Search stops working.

D. The value of linear_index becomes -1.

Answer: B

Justification:

Without the break, the loop continues until the end of the list. Since the target appears only once, the final value of linear_index remains unchanged.

Q5. (MSQ)

Which of the following statements are correct?

A. Linear Search checks elements one by one from the beginning of the list.

B. Binary Search repeatedly divides the search space into two halves.

C. Both Linear Search and Binary Search return index 4 for the given input.

D. Binary Search can be applied correctly even if book_ids is not sorted.

Answer: A, B, C

Justification:

Linear Search examines elements sequentially, while Binary Search repeatedly halves the search range. Since the target exists at index 4, both algorithms return 4; however, Binary Search requires the list to be sorted.

Scenario: Airport Boarding Gate Search

An airport maintains a sorted list of boarding gate numbers for departing flights.

[101, 104, 108, 112, 117, 121, 126, 130, 135, 140]

A passenger is looking for Gate 126. The airport system can use either Linear Search or Binary Search to locate the gate.

Answer the following questions.

Q1. (MCQ)

How many elements would Linear Search examine before finding Gate 126?

A. 5

B. 6

C. 7

D. 8

Answer: C

Justification:
Linear Search checks elements one by one from the beginning. Gate 126 is the 7th element in the list, so seven comparisons are required.

Q2. (MCQ)
During the first comparison of Binary Search, which gate number is examined?
A. 112
B. 117
C. 121
D. 126
Answer: B
Justification:
The middle index of a 10-element list is (0 + 9) // 2 = 4. The element at index 4 is 117.

Q3. (MCQ)
After the first Binary Search comparison, which part of the list will be searched next?
A. [101, 104, 108, 112]
B. [101, 104, 108, 112, 117]
C. [121, 126, 130, 135, 140]
D. [126, 130, 135, 140]
Answer: C
Justification:
Since 126 is greater than 117, Binary Search discards the left half. It continues searching the right half of the list.

Q4. (MCQ)
Why is Binary Search generally faster than Linear Search for large sorted lists?
A. It compares every element only once.
B. It checks two elements in every comparison.
C. It eliminates half of the remaining search space after each comparison.
D. It works only on lists with unique values.
Answer: C

Justification:

Binary Search reduces the search space by half after each comparison. This makes it much more efficient than checking elements one by one.

Q5. (MSQ)

Which of the following statements are correct?

A. Binary Search requires the list to be sorted.

B. Linear Search can be used on both sorted and unsorted lists.

C. Both Linear Search and Binary Search will locate Gate 126 in this scenario.

D. Binary Search always examines every element in the list.

Answer: A, B, C

Justification:

Binary Search works correctly only on sorted data, whereas Linear Search works on any list. In this scenario, both methods successfully locate Gate 126, but Binary Search does not examine every element.

Week 06 – Assignment

Scenario: Secret Notice Encoder

A college club regularly shares announcements with its members. Since these messages are posted on a public notice board, the club wants to make them difficult for outsiders to read. To achieve this, the club uses a Substitution Cipher. Every alphabetic character is replaced by the previous letter in string.ascii_letters. Non-alphabetic characters such as spaces, digits, and punctuation marks remain unchanged.

The following Python program performs this encryption.

```
1. import string
2.
3. cipher = {}
4.
5. letters = string.ascii_letters
6.
7. for i in range(len(letters)):
8.     cipher[letters[i]] = letters[i - 1]
9.
10. text = input("Enter a message: ")
11.
12. encrypted = ""
13.
14. for ch in text:
15.     if ch in cipher:
16.         encrypted += cipher[ch]
17.     else:
18.         encrypted += ch
19.
20. print("Original :", text)
21. print("Encrypted:", encrypted)
```

Q1. (MCQ)

Which line creates the substitution rule that maps each letter to its replacement?

A. Line 5

B. Line 7

C. Line 8

D. Line 16

Answer: C

Justification: Line 8 stores each original letter as a key and its substituted letter as the corresponding value in the dictionary.

Q2. (MCQ)

Suppose the user enters the input:

ABC123

Which statement ensures that the digits 1, 2, and 3 remain unchanged?

A. Line 10

B. Lines 17–18

C. Line 20

D. Line 21

Answer: B

Justification: If a character is not found in the dictionary (such as digits or punctuation), the else block copies it unchanged.

Q3. (MCQ)

What is the purpose of Line 12?

A. It stores the substitution dictionary.

B. It stores the original input.

C. It initializes an empty string to build the encrypted message.

D. It prints the encrypted output.

Answer: C

Justification: The variable encrypted starts as an empty string, and encrypted characters are appended to it during processing.

Q4. (MCQ)

If Line 16

encrypted += cipher[ch]

is replaced with

encrypted += ch

what will be the result?

A. Only uppercase letters will be encrypted.

B. The program will generate an error.

C. Only digits will be encrypted.

D. The output will be identical to the input.

Answer: D

Justification: The substitution dictionary is no longer used. Every character is copied directly into the output.

Q5. (MSQ)

Which of the following statements about the program are correct? (Select all that apply.)

A. Line 5 stores all lowercase and uppercase English letters.

B. Line 8 creates a substitution mapping using a dictionary.

C. Line 15 checks whether the current character is a digit.

D. Lines 20 and 21 display both the original and encrypted messages.

Answer: A, B, D

Justification:

- A: string.ascii_letters contains both lowercase and uppercase English letters.
- B: Line 8 creates the mapping from each letter to its substitute.
- C: Line 15 checks whether the character exists in the dictionary, not whether it is a digit.
- D: The final two print statements display both the original input and the encrypted output.

Scenario: Tic-Tac-Toe Winner Checker

A programming club is developing a simple Tic-Tac-Toe game. The game board is represented using a nested list, where each inner list represents a row of the board.

- 'X' represents Player X.
- 'O' represents Player O.
- ' ' (space) represents an empty cell.

The program checks only the rows of the board to determine whether either player has won. If a row contains three identical non-space symbols, that player is declared the winner. Otherwise, the program reports that there is no row winner.

```
1. board = [
2.     ['X', 'X', 'X'],
3.     ['O', ' ', 'O'],
4.     [' ', 'O', ' ']
5. ]
6.
7. winner = "None"
8.
9. for row in board:
10.     if row[0] == row[1] == row[2] and row[0] != ' ':
11.         winner = row[0]
12.         break
13.
14. if winner != "None":
15.     print("Winner:", winner)
16. else:
17.     print("No row winner")
```

Q1. (MCQ)

Which line checks whether all three cells in a row contain the same non-empty symbol?

A. Line 7

B. Line 9

C. Line 10

D. Line 15

Answer: C

Justification: Line 10 compares all three elements of a row and ensures that they are not blank.

Q2. (MCQ)

What is the purpose of the break statement in Line 12?

A. It skips the current row.

B. It exits the loop after finding a winner.

C. It resets the board.

D. It starts checking the rows again.

Answer: B

Justification: Once a winning row is found, there is no need to examine the remaining rows.

Q3. (MCQ)

Which variable stores the winning player's symbol?

A. board

B. row

C. winner

D. print

Answer: C

Justification: The variable winner is updated with 'X' or 'O' when a winning row is found.

Q4. (MCQ)

If Line 11

winner = row[0]

is replaced with

winner = "Player"

what will the program display when a winning row is found?

A. The winning symbol (X or O)

B. "Player"

C. "None"

D. The entire row

Answer: B

Justification: The variable winner would always contain the string "Player" instead of the winning symbol.

Q5. (MSQ)

Which of the following statements about the program are correct? (Select all that apply.)

A. The board is represented using a nested list.

B. The program checks only row-wise winning conditions.

C. The loop stops immediately after a winner is found.

D. The program checks both row-wise and diagonal winning conditions.

Answer: A, B, C

Justification:

- A: Correct. The board is stored as a list of lists.
- B: Correct. The program examines only the rows for a winning condition.
- C: Correct. The break statement terminates the loop once a winner is detected.

- D: Incorrect. The program does not check diagonal or column-wise winning conditions.

Scenario-Based Question: Recursive Folder Search

A company stores its documents in a hierarchy of folders. Each folder can contain:

- Files
- Other folders

A software engineer is asked to write a program to find a file named Report.pdf. The program begins with the main folder. Whenever it encounters a subfolder, it performs the same search process inside that subfolder. If the file is found, the search stops. If a folder contains no more subfolders or files, the program returns to the previous folder and continues searching.
No loops are explicitly used to revisit folders; instead, the same search procedure is repeatedly applied to each subfolder.

Questions

Q1. (MCQ)

Which characteristic of the search process best represents recursion?

A. The search visits folders in alphabetical order.

B. The same search procedure is applied repeatedly to each subfolder.

C. Every folder contains the same number of files.

D. The search uses multiple computers simultaneously.

Answer: B

Justification: In recursion, the same task is performed repeatedly on smaller instances of the same problem.

Q2. (MCQ)

In this scenario, what is the base case for the recursive search?

A. The search begins from the main folder.

B. A folder contains only PDF files.

C. A folder has no more files or subfolders to search, or the required file is found.

D. The search enters a subfolder.

Answer: C

Justification: A recursive process stops when it reaches a condition where no further recursive calls are needed.

Q3. (MCQ)

Why is recursion a suitable approach for this problem?

A. Every folder has exactly one subfolder.

B. The same search operation must be repeated for each subfolder.

C. Recursion is always faster than iteration.

D. Files can only exist in the main folder.

Answer: B

Justification: Each subfolder is simply a smaller version of the original problem, making recursion a natural solution.

Q4. (MCQ)

What would most likely happen if the search process had no base case?

A. The search would immediately stop.

B. The search would examine only the main folder.

C. The search could continue indefinitely or until system resources are exhausted.

D. The search would automatically switch to an iterative approach.

Answer: C

Justification: Without a stopping condition, recursive calls continue indefinitely, eventually causing a stack overflow or exhausting system resources.

Q5. (MSQ)

Which of the following statements correctly describe recursion in the given scenario? (Select all that apply.)

A. The same search procedure is used for every subfolder.

B. The search must have a stopping condition.

C. Each subfolder represents a smaller instance of the original problem.

D. Recursion can only be used if every folder contains exactly one subfolder.

Answer: A, B, C

Justification:

- A: Correct. The same search logic is applied to every subfolder.
- B: Correct. Every recursive solution requires a base case to terminate.
- C: Correct. Each subfolder is a smaller version of the original folder search problem.

- D: Incorrect. A folder may contain any number of subfolders or none at all; recursion does not require exactly one subfolder.

Scenario: Recursive Binary Search for Student Roll Numbers

A university stores student roll numbers in ascending order. When an administrator wants to verify whether a particular roll number exists, the system performs a recursive binary search. The search begins by checking the middle roll number in the current range.

- If the middle roll number matches the required roll number, the search is successful.
- If the required roll number is smaller, the same search process is repeated on the left half.
- If the required roll number is larger, the search continues recursively on the right half.
- If the search range becomes invalid (the left index becomes greater than the right index), the roll number is reported as not found.

The following program implements this recursive binary search.

```
1. def binary_search(arr, low, high, target):
2.     if low > high:
3.         return -1
4.
5.     mid = (low + high) // 2
6.
7.     if arr[mid] == target:
8.         return mid
9.
10.    elif target < arr[mid]:
11.        return binary_search(arr, low, mid - 1, target)
12.
13.    else:
14.        return binary_search(arr, mid + 1, high, target)
15.
16. numbers = [105, 112, 118, 124, 131, 145, 152, 168]
17. key = 131
18.
19. result = binary_search(numbers, 0, len(numbers)-1, key)
20.
21. if result != -1:
22.     print("Roll number found at index", result)
23. else:
24.     print("Roll number not found")
```

Q1. (MCQ)

Which line represents the base case where the search terminates because the search range has become invalid?

A. Line 5

B. Line 2

C. Line 7

D. Line 14

Answer: B

Justification: Line 2 checks whether low > high. If true, there are no more elements to search, and the recursion terminates.

Q2. (MCQ)

Which line performs the recursive search on the left half of the list?

A. Line 5

B. Line 8

C. Line 11

D. Line 19

Answer: C

Justification: When the target is smaller than the middle element, the function recursively searches from low to mid - 1.

Q3. (MCQ)

What is the purpose of Line 5?

A. It checks whether the element is found.

B. It calculates the middle index of the current search range.

C. It sorts the list.

D. It prints the search result.

Answer: B

Justification: The middle index is calculated so that the current search range can be divided into two halves.

Q4. (MCQ)

If Line 14

return binary_search(arr, mid + 1, high, target)
is replaced with
return binary_search(arr, low, high, target)
what is the most likely outcome when the target is greater than the middle element?
A. The program always finds the target immediately.
B. The search correctly moves to the right half.
C. The search continues without reducing the search range, leading to infinite recursion.
D. The program searches only the left half.
Answer: C
Justification: Since the search boundaries do not change, the same recursive call is repeated indefinitely.

Q5. (MSQ)
Which of the following statements about the program are correct? (Select all that apply.)
A. The recursive calls in Lines 11 and 14 reduce the search space in every recursive step.
B. The program can correctly perform binary search even if numbers is not sorted.
C. Line 5 compares the target with every element in the list.
D. The recursion stops either when the target is found (Line 8) or when the search range becomes invalid (Line 3).
Answer: A, D
Justification:

- A: Correct. Each recursive call searches only one half of the remaining list.
- B: Incorrect. Binary search requires the list to be sorted.
- C: Incorrect. Line 5 only computes the middle index; it does not compare every element.
- D: Correct. These are the two terminating conditions of the recursive binary search.

Week 07 – Assignments

Scenario: Drone Surveillance Route

A disaster management team uses a drone to capture images of a disaster-affected area. The images are stored in a matrix. During analysis, the system processes the images in a spiral order, beginning from the outer boundary and gradually moving toward the center so that all regions are examined systematically.

The following Python program performs the spiral traversal of the matrix.

```
def spiral(m, n, a):
    k = 0
    l = 0

    while k < m and l < n:

        for i in range(l, n):
            print(a[k][i], end=" ")
        k += 1

        for i in range(k, m):
            print(a[i][n-1], end=" ")
        n -= 1

        if k < m:
            for i in range(n-1, l-1, -1):
                print(a[m-1][i], end=" ")
            m -= 1

        if l < n:
            for i in range(m-1, k-1, -1):
                print(a[i][l], end=" ")
            l += 1

matrix = [
    [1, 2, 3, 4],
    [5, 6, 7, 8],
    [9, 10, 11, 12],
    [13, 14, 15, 16]
]

spiral(4, 4, matrix)
```

Question 1 (MCQ)

Which variables store the starting row and starting column of the current spiral layer?

A. k and l

B. m and n

C. i and k

D. a and i

Correct Answer: A

Justification: k stores the starting row index and l stores the starting column index.

Question 2 (MCQ)

What is the purpose of the statement at line 9?

A. It moves to the next column.

B. It moves to the next row after printing the top row.

C. It moves to the previous row.

D. It resets the row index.

Correct Answer: B

Justification: The starting row index is increased after the top row has been printed.

Question 3 (MCQ)

Which statement best describes the loop at lines 11–12?

A. It prints the first column from top to bottom.

B. It prints the last row from left to right.

C. It prints the last column from top to bottom.

D. It prints the top row from right to left.

Correct Answer: C

Justification: The column index remains fixed at n-1 while the row index increases.

Question 4 (MCQ)

What is the first sequence of numbers printed by the program?

A. 1 2 3 4

B. 4 8 12 16

C. 13 14 15 16

D. 1 5 9 13

Correct Answer: A

Justification: The algorithm begins by traversing the top row from left to right.

Question 5 (MSQ)

Which of the following statements are true about the program?

A. The variables k, l, m, and n define the current boundaries of the unvisited portion of the matrix.
B. After each side of the spiral is printed, the corresponding boundary is updated.
C. The program prints only square matrices correctly.
D. The program traverses the matrix column by column throughout.
Correct Answers: A, B
Justification: The algorithm maintains and updates four boundaries to progressively traverse the matrix in spiral order.

Scenario: Mini Snake and Ladder Tournament

A board game club is testing a simplified Snake and Ladder game. The board has 25 positions. A player starts at position 1 and moves according to a sequence of dice rolls. If the player lands on the bottom of a ladder, the player climbs immediately. If the player lands on the head of a snake, the player slides down immediately. If the player's position becomes greater than 25, the move is ignored.
The game has the following snakes and ladders:

- Ladder: 4 → 12
- Ladder: 9 → 18
- Snake: 16 → 7
- Snake: 22 → 14

The following Python program simulates the player's journey.

```
def play_game():
    position = 1
    target = 25
    rolls = [3, 5, 7, 6, 4, 2]

    for roll in rolls:
        if position + roll <= target:
            position += roll

        if position == 4:
            position = 12
        elif position == 9:
            position = 18
        elif position == 16:
            position = 7
        elif position == 22:
            position = 14

        print(position, end=" ")

        if position == target:
            break

play_game()
```

Question 1 (MCQ)

Which statement ensures that the player does not move beyond the last position on the board?

A. if position == target:

B. if position + roll <= target:

C. position += roll

D. break

Correct Answer: B

Justification: The condition allows a move only if the new position does not exceed the target.

Question 2 (MCQ)

If the player reaches position 16, what happens next?

A. The player moves to position 22.

B. The player stays at position 16.

C. The player moves to position 18.

D. The player moves to position 7.

Correct Answer: D

Justification: Position 16 is the head of a snake that sends the player to position 7.

Question 3 (MCQ)

What is the purpose of the statement at line 21?

A. It checks whether all dice rolls are completed.

B. It checks whether the player has reached the target position.

C. It resets the player's position.

D. It moves the player to the next turn.

Correct Answer: B

Justification: The condition checks if the player has reached the final position.

Question 4 (MCQ)

Which statement immediately ends the loop once the player reaches the target?

A. break

B. position += roll

C. print(position, end=" ")

D. for roll in rolls:

Correct Answer: A

Justification: The break statement terminates the loop immediately.

Question 5 (MSQ)

Which of the following statements are true about the program?

A. The player always moves after every dice roll.

B. The game stops only after all dice rolls are processed.

C. A move is ignored if it takes the player beyond position 25.

D. Landing on a snake or ladder changes the player's position immediately.

Correct Answers: C, D

Justification: The program ignores moves beyond the target and immediately applies snakes or ladders whenever the player lands on them.

Scenario: Food Delivery Pickup Locations

A food delivery company uses a map to display the locations of restaurants where delivery agents need to collect orders. Each restaurant is represented by a marker on the map. When a marker is clicked, the restaurant's name is displayed.

The following Python program creates a map and places markers for three restaurants.

```
1   import gmplot
2
3   gmap = gmplot.GoogleMapPlotter(15.3647, 75.1240, 13)
4
5   restaurants = [
6       ("Pizza Hub", 15.3655, 75.1258),
7       ("Green Bowl", 15.3628, 75.1219),
8       ("Spice Corner", 15.3672, 75.1285)
9   ]
10
11  for name, lat, lon in restaurants:
12      gmap.marker(lat, lon, title=name)
13
14  gmap.draw("restaurants.html")
```

Question 1 (MCQ)

What is the purpose of the statement at line 3?

A. It creates a marker on the map.

B. It stores restaurant names.

C. It creates a Google map centered at the given location.

D. It saves the map as an HTML file.

Correct Answer: C

Justification: The statement creates a Google Map object centered at the specified latitude and longitude.

Question 2 (MCQ)

Which statement adds a marker for each restaurant?

A. gmap.draw("restaurants.html")

B. gmap.marker(lat, lon, title=name)

C. restaurants = [...]

D. for name, lat, lon in restaurants:

Correct Answer: B

Justification: The marker() method places a marker at the specified coordinates.

Question 3 (MCQ)

What is displayed as the title of each marker?

A. The latitude value

B. The longitude value

C. The map's zoom level

D. The restaurant name

Correct Answer: D

Justification: The value of name is passed as the marker's title.

Question 4 (MCQ)

What is the purpose of the statement at line 14?

A. It creates another marker.

B. It removes all markers from the map.

C. It changes the map center.

D. It saves the map as an HTML file.

Correct Answer: D

Justification: The draw() method generates the HTML file containing the map.

Question 5 (MSQ)

Which of the following statements are true about the program?

A. The program creates a separate map for each restaurant.

B. The markers are added using the latitude and longitude values.

C. The program iterates through all restaurants using a for loop.

D. The HTML file is created before the markers are added.

Correct Answers: B, C

Justification: The loop processes each restaurant and marker() uses its latitude and longitude to place a marker on the map.

Case Study: Smart Heritage City Explorer

The tourism department of a heritage city has developed a Smart Heritage City Explorer to help visitors explore popular attractions.

The city is represented as a matrix, where each cell corresponds to a location. Every morning, an inspection drone scans the city in a spiral order, starting from the outer boundary and moving towards the center to ensure that every location is inspected exactly once.

Information about each attraction is stored in a dictionary containing the attraction's name, category, visitor rating, latitude, and longitude. When a visitor searches for an attraction, the

application uses linear search if the attractions are stored in random order. If the attractions are sorted alphabetically, the application uses binary search to locate the attraction more efficiently.

After an attraction is found, its latitude and longitude are used to place a marker on a Google Map so that visitors can easily navigate to the destination.

The application also includes a Treasure Trail game. Visitors move across 25 checkpoints using dice rolls. Some checkpoints contain ladders that move visitors ahead, while others contain snakes that move them backward. If a dice roll takes a visitor beyond the final checkpoint, that move is ignored.

The developers use appropriate data structures and algorithms so that the application remains efficient and easy to maintain.

Question 1 (MCQ)

The attractions are stored in random order. Which searching technique is most appropriate for finding a particular attraction?

A. Binary Search

B. Spiral Search

C. Linear Search

D. Depth First Search

Correct Answer: C

Justification: Linear search is suitable when the data is not sorted.

Question 2 (MCQ)

Why does the inspection drone use a spiral traversal of the city matrix?

A. To visit only the corner locations.

B. To inspect every location by moving from the outer boundary towards the center.

C. To sort the locations alphabetically.

D. To search for a specific attraction.

Correct Answer: B

Justification: Spiral traversal systematically covers all cells from the outer layer to the center.

Question 3 (MCQ)

Which information is essential for placing an attraction on the Google Map?

A. Name and Category

B. Rating and Category

C. Latitude and Longitude

D. Name and Rating

Correct Answer: C

Justification: A map marker requires geographic coordinates to determine its location.

Question 4 (MCQ)

In the Treasure Trail game, what should happen if a dice roll moves the visitor beyond checkpoint 25?

A. The visitor returns to checkpoint 1.

B. The visitor wins immediately.

C. The visitor moves to checkpoint 25.

D. The move is ignored.

Correct Answer: D

Justification: The game rules specify that moves beyond the final checkpoint are not allowed.

Question 5 (MSQ)

Which of the following statements are true about the Smart Heritage City Explorer?

A. Binary search can be used efficiently even if the attractions are stored in random order.

B. A dictionary is suitable for storing multiple details about each attraction.

C. Spiral traversal is a suitable approach for visiting every location in a matrix.

D. Google Map markers are placed using the latitude and longitude of an attraction.

Correct Answers: B, C

Justification: Dictionaries can store multiple attributes of an attraction, and spiral traversal visits every matrix location exactly once.

Week 08 – Assignment

Scenario: Student Record Management

A college stores each student's information as a tuple because these details do not change during a semester.

One student's record is stored as:

("Ananya", 1023, "Computer Science", 8.7)

The information is stored in the following order:

- Student Name
- Student ID
- Department
- CGPA

The college uses this tuple whenever it needs to retrieve a student's information.

Q1. Which piece of information is stored in the third position of the tuple?

A. Student Name

B. Student ID

C. Department

D. CGPA

Correct Answer: C

Justification: The third value in the tuple is "Computer Science", which represents the student's department.

Q2. Why is a tuple a suitable data structure for storing this record?

A. Because the student details are not expected to change frequently.

B. Because tuples automatically sort the data.

C. Because tuples allow duplicate student IDs.

D. Because tuples can only store text values.

Correct Answer: A

Justification: Tuples are useful when the stored information remains fixed after creation.

Q3. How many pieces of information are stored in the student's tuple?

A. 2

B. 3

C. 4

D. 5

Correct Answer: C

Justification: The tuple contains four values: Name, Student ID, Department, and CGPA.

Q4. Which of the following is the last item stored in the tuple?

A. Ananya

B. 1023

C. Computer Science

D. 8.7

Correct Answer: D

Justification: The last value in the tuple is the student's CGPA, which is 8.7.

Q5. A staff member wants to replace the department "Computer Science" with "Artificial Intelligence" in the existing tuple. What should they do?

A. Modify the existing tuple directly.

B. Create a new tuple with the updated value.

C. Delete only the department value.

D. Use the append() operation.

Correct Answer: B

Justification: Tuples are immutable, so their values cannot be modified after creation. A new tuple must be created with the updated information.

Scenario: Lucky Draw Betting Game

A gaming café has introduced a simple Lucky Draw Challenge. A player starts with ₹0 in their game account.

The game is played 7 times. In every round:

- The player's bet number is randomly chosen between 1 and 10.
- The lucky draw number is also randomly chosen between 1 and 10.
- If both numbers are the same, the player wins ₹900, but ₹100 is deducted as the participation fee. Hence, the account increases by ₹800.
- If the numbers are different, the player only loses the ₹100 participation fee.

- After every round, the current account balance is displayed.

The following Python program implements this game.

```
1. import random
2.
3. account = 0
4.
5. for i in range(7):
6.     bet = random.randint(1, 10)
7.     lucky_draw = random.randint(1, 10)
8.     print("Bet:", bet)
9.     print("Lucky draw:", lucky_draw)
10.
11.    if bet == lucky_draw:
12.        account = account + 900 - 100
13.    else:
14.        account = account - 100
15.
16.    print("Amount in your game account:", account)
```

Q1. According to Line 5, how many rounds will the game be played?

A. 7

B. 8

C. 10

D. Depends on the random numbers

Correct Answer: A

Justification: range(7) repeats the loop 7 times, so the game consists of seven rounds.

Q2. Which lines generate the random numbers for the player's bet and the lucky draw?

A. Lines 3 and 5

B. Lines 8 and 9

C. Lines 11 and 12

D. Lines 6 and 7

Correct Answer: D

Justification: Lines 6 and 7 use random.randint(1, 10) to generate the bet number and the lucky draw number.

Q3. If the bet number and lucky draw number are equal, what is the net change in the account balance according to Line 12?

A. +₹100

B. +₹900

C. +₹700

D. +₹800

Correct Answer: D

Justification: The player receives ₹900 and pays a ₹100 participation fee. Therefore, the net increase is ₹800.

Q4. According to Lines 13–14, what happens when the bet number and lucky draw number are different?

A. ₹100 is deducted from the account.

B. ₹900 is added to the account.

C. The account balance remains unchanged.

D. A new account is created.

Correct Answer: A

Justification: The else block subtracts ₹100 from the account, representing the participation fee.

Q5. (MSQ) Which of the following statements about the program are correct? *(Choose all that apply.)*

A. The player's account balance is printed after every round.

B. The player's bet is entered by the user using input().

C. The lucky draw number is always different from the bet number.

D. The player's account balance can become negative.

Correct Answers: A and D

Justification:

- A Line 16 prints the account balance after each iteration of the loop.
- B The program generates the bet randomly; it does not use input().
- C Since both values are random, they may or may not be equal.
- D If the player loses multiple rounds, ₹100 is deducted each time, so the account balance can become negative.

Scenario: Secure Access Puzzle Generator

A cybersecurity training platform uses anagrams to create authentication puzzles. Before granting access to a practice lab, the system displays a shuffled version of a secret keyword. To ensure the puzzle is meaningful, the displayed word must not be identical to the original keyword. The system keeps generating new arrangements until a valid anagram is produced. The following Python program implements this logic.

```
1. import random
2.
3. keyword = input("Enter the secret keyword: ")
4. letters = list(keyword)
5.
6. while True:
7.     random.shuffle(letters)
8.     candidate = "".join(letters)
9.
10.    if candidate != keyword:
11.        break
12.
13. print("Authentication Puzzle:", candidate)
```

Q1. Suppose the user enters the word LEVEL. Which of the following statements is always true?

A. The displayed puzzle will never contain repeated letters.

B. The displayed puzzle will always have exactly five letters.

C. The displayed puzzle will always be alphabetically sorted.

D. The displayed puzzle cannot be generated because the word contains repeated letters.

Correct Answer: B

Justification: Shuffling changes only the order of characters. The number of characters always remains the same. Repeated letters do not prevent an anagram from being generated.

Q2. Consider the execution of Lines 6–11. Which situation can cause the loop to execute more than once?

A. The entered keyword has fewer than five letters.

B. random.shuffle() arranges the letters in the same order as the original keyword.

C. The keyword contains only uppercase letters.

D. The keyword contains duplicate characters.

Correct Answer: B

Justification: If the shuffled arrangement is identical to the original word, the condition on Line 10 fails and the loop repeats until a different arrangement is obtained.

Q3. Which of the following inputs may cause the program to never terminate?

A. PYTHON

B. RANDOM

C. A

D. SECURE

Correct Answer: C

Justification: A single-character word has only one possible arrangement. Therefore, candidate will always be equal to keyword, making the condition on Line 10 false forever.

Q4. Which statement best describes the purpose of Lines 10–11?

A. They ensure that every possible anagram is generated exactly once.

B. They ensure that duplicate letters are removed.

C. They stop the loop only after a shuffled word different from the original has been generated.

D. They guarantee that the generated anagram is alphabetically smallest.

Correct Answer: C

Justification: The loop terminates only when the generated arrangement differs from the original word.

Q5. (MSQ) Which of the following statements about the program are correct? (Choose all that apply.)

A. If the input contains only one character, the program may run indefinitely.

B. Every execution of the program produces a different anagram.

C. The program guarantees that all possible anagrams will eventually be generated.

D. The generated puzzle always contains exactly the same characters as the original keyword.

Correct Answers: A and D

Justification:

- A Correct. A one-character word has only one possible arrangement, so the loop never satisfies the condition on Line 10.

- B Incorrect. Randomness does not guarantee a different output on every execution. The same anagram may be generated in different runs.
- C Incorrect. The program stops after generating the first valid anagram. It does not enumerate all possible anagrams.
- D Correct. random.shuffle() only changes the order of the existing characters; it does not add, remove, or replace any characters.

Scenario: Customer Feedback Analysis

An online shopping company collects customer reviews after every purchase. The data is stored in a Pandas DataFrame. To understand customer satisfaction, the company uses NLTK's VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment analyzer. Each review is assigned a compound sentiment score. Reviews with a positive compound score are labeled Positive; otherwise, they are labeled Negative.

The following Python program performs this analysis.

```
1. import pandas as pd
2. from nltk.sentiment.vader import SentimentIntensityAnalyzer
3.
4. reviews = pd.DataFrame({
5.     "Customer": ["Asha", "Rahul", "Neha", "Arjun"],
6.     "Review": [
7.         "Excellent product and fast delivery",
8.         "The quality is average",
9.         "Very disappointed with the service",
10.        "Worth every penny"
11.    ]
12. })
13.
14. analyzer = SentimentIntensityAnalyzer()
15.
16. def get_sentiment(review):
17.     score = analyzer.polarity_scores(review)["compound"]
18.     if score > 0:
19.         return "Positive"
20.     else:
21.         return "Negative"
22.
23. reviews["Sentiment"] = reviews["Review"].apply(get_sentiment)
24.
25. print(reviews)
```

Q1. What is the primary purpose of Line 14?

A. To create a Pandas DataFrame.

B. To create an object that can analyze the sentiment of text.

C. To read customer reviews from a file.

D. To display the final DataFrame.

Correct Answer: B

Justification: SentimentIntensityAnalyzer() creates a VADER sentiment analyzer that is used to calculate sentiment scores for text.

Q2. According to Line 17, which value is used to determine whether a review is positive or negative?

A. neg

B. compound

C. neu

D. pos

Correct Answer: B

Justification: The program extracts the compound score from the dictionary returned by polarity_scores(). This score is used in the decision on Lines 18–21.

Q3. What is the purpose of Line 23?

A. It removes reviews with negative sentiment.

B. It creates a new column containing the sentiment label for each review.

C. It sorts the reviews based on sentiment.

D. It replaces the original review text with sentiment scores.

Correct Answer: B

Justification: The apply() function calls get_sentiment() for every review and stores the returned labels in the new Sentiment column.

Q4. Which statement best describes the role of the function get_sentiment()?

A. It converts reviews into uppercase.

B. It counts the number of words in each review.

C. It calculates the average sentiment score of all reviews.

D. It assigns a sentiment label to a single review based on its compound score.

Correct Answer: D

Justification: The function computes the compound score for one review and returns either "Positive" or "Negative".

Q5. (MSQ) Which of the following statements are correct? (Choose all that apply.)

A. The program stores the sentiment score directly in the Review column.

B. The reviews are stored in a Pandas DataFrame.

C. The sentiment label for each review is generated using the get_sentiment() function.

D. The program classifies reviews based on the length of the review text.

Correct Answers: B and C

Justification:

- A. Incorrect. The original Review column is not modified.
- B. Correct. Lines 4–12 create a Pandas DataFrame named reviews.
- C. Correct. Line 23 applies the get_sentiment() function to every review.
- D. Incorrect. Classification is based on the compound sentiment score, not the length of the review.

Week 09 – Assignment

Scenario: Product Review Analysis

An online marketplace wants to analyze the words used in customer reviews. The program tokenizes the review text, creates a frequency distribution using FreqDist, and displays only the words that occur more than once. It also reports the total number of unique words.

```
1. from nltk.tokenize import word_tokenize
2. from nltk import FreqDist
3.
4. reviews = """
5. Fast delivery and good packaging.
6. Good quality and fast service.
7. Delivery was fast and packaging was good.
8. """
9.
10. tokens = word_tokenize(reviews.lower())
11. freq = FreqDist(tokens)
12.
13. print(len(tokens))
14. print(len(freq))
15.
16. for word in sorted(freq):
17.     if freq[word] > 1:
18.         print(word, freq[word])
```

Q1. Which statement converts the review text into a list of tokens?

A. Line 10

B. Line 11

C. Line 13

D. Line 16

Answer: A

Justification: word_tokenize() splits the given text into individual tokens and stores them in the list tokens.

Q2. What does the expression len(freq) in Line 14 return?

A. Total number of characters in the review

B. Number of unique tokens in the review

C. Total number of repeated words

D. Number of lines in the review

Answer: B

Justification: A FreqDist object stores one entry for each distinct token, so len(freq) returns the number of unique tokens.

Q3. Which line creates the frequency distribution of the tokens?

A. Line 10

B. Line 16

C. Line 11

D. Line 13

Answer: C

Justification: FreqDist(tokens) counts the occurrences of every token and stores the resulting frequency distribution.

Q4. Which part of the program prints only the words that occur more than once?

A. Line 10

B. Line 13

C. Line 14

D. Lines 16–18

Answer: D

Justification: The loop checks each token's frequency and prints only those whose count is greater than 1.

Q5. (MSQ) Which of the following statements are correct?

A. len(tokens) returns the number of unique words in the review.

B. word_tokenize() automatically counts the frequency of each token.

C. FreqDist stores each token as a key and its frequency as the corresponding value.

D. len(freq) gives the number of distinct tokens present in the text.

Answer: C and D

Justification: FreqDist behaves like a dictionary with tokens as keys and counts as values, and len(freq) returns the number of distinct keys stored in the distribution.

Scenario: Library Book Recommendation Network

A digital library recommends books based on their similarity. Each book is represented as a node, and an edge between two books indicates that they are frequently borrowed together. The librarian uses NetworkX to model and examine this recommendation network.

```
import networkx as nx

G = nx.Graph()

G.add_nodes_from(["Python", "AI", "ML", "Data", "Cloud"])

G.add_edges_from([
    ("Python", "AI"),
    ("Python", "Data"),
    ("AI", "ML"),
    ("ML", "Data"),
    ("Cloud", "Data")
])

print(len(G.nodes))
print(len(G.edges))

for book in G.nodes:
    print(book, list(G.neighbors(book)))
```

Q1. In this graph, what does a node represent?

A. A book

B. A recommendation between two books

C. The total number of books

D. A borrowing record

Answer: A

Justification: Each node represents one book in the recommendation network.

Q2. What does an edge between two books indicate?

A. The books have the same author.

B. The books belong to different categories.

C. The books are borrowed together and are recommended to each other.

D. The books have the same publication year.

Answer: C

Justification: Each edge represents a recommendation relationship between two books.

Q3. What does len(G.nodes) represent?

A. Number of recommendations

B. Number of books in the network

C. Number of neighboring books

D. Number of connected components

Answer: B

Justification: G.nodes contains all the books (nodes) present in the graph.

Q4. Why is G.neighbors(book) used in the program?

A. To remove duplicate books

B. To sort the books alphabetically

C. To calculate the shortest path

D. To find books directly connected to a given book

Answer: D

Justification: neighbors() returns all nodes that share an edge with the given node.

Q5. (MSQ) Which of the following statements are correct?

A. Every node must be connected to every other node.

B. An edge can exist without any nodes.

C. len(G.edges) returns the total number of recommendation links in the graph.

D. A graph can be used to model relationships between connected objects.

Answer: C and D

Justification: Edges represent relationships between nodes, and graphs are commonly used to model such connections.

Scenario: Six Degrees of Separation

A social networking platform represents its users as a graph. Each person is a node, and a friendship between two people is represented by an edge. According to the "Six Degrees of Separation" concept, any two people in a social network can usually be connected through a small chain of mutual friends. The platform analyzes these connections to recommend new friends and understand how closely users are connected.

Q1. In the social network described above, what does an edge represent?

A. A user profile

B. A friendship between two users

C. The total number of users

D. A friend recommendation

Answer: B

Justification: An edge represents a direct friendship connecting two users.

Q2. Alice is friends with Bob, and Bob is friends with Carol. Alice and Carol are not directly connected. What exists between Alice and Carol?

A. A path through Bob

B. A duplicate edge

C. A disconnected graph

D. A self-loop

Answer: A

Justification: A path connects two nodes through one or more intermediate nodes.

Q3. Why is the shortest path between two users often important in a social network?

A. It increases the total number of users.

B. It removes duplicate friendships.

C. It identifies the minimum number of friendship links connecting two users.

D. It converts the graph into a directed graph.

Answer: C

Justification: The shortest path represents the fewest intermediate connections between two users.

Q4. When is a social network considered to be connected?

A. Every user has exactly one friend.

B. Every user is directly connected to every other user.

C. The graph has no edges.

D. A path exists between every pair of users.

Answer: D

Justification: A connected graph allows every node to be reached from every other node through at least one path.

Q5. (MSQ) Which of the following statements are correct?

A. Every pair of users must have a direct friendship for a path to exist.

B. A graph cannot have more than one path between two users.

C. A node represents an individual user in the social network.

D. The Six Degrees of Separation concept is based on finding paths between people in a network.

Answer: C and D

Justification: In a social graph, nodes represent users, and the Six Degrees concept relies on paths that connect people through mutual friends.

Scenario: Quality Inspection of Product Images

An online retail company inspects product images before publishing them on its website. The system opens an image, checks the color of a specific pixel, marks that pixel with a new color for verification, creates a small thumbnail image, and saves the modified image.

```
1. from PIL import Image
2.
3. img = Image.open("product.jpg")
4.
5. print(img.size)
6. print(img.getpixel((20, 20)))
7.
8. img.putpixel((20, 20), (255, 0, 0))
9.
10. thumb = Image.new("RGB", (100, 100), "white")
11. thumb.save("thumbnail.jpg")
12.
13. img.save("verified_product.jpg")
```

Q1. Why is getpixel() used in this program?

A. To resize the image

B. To change the image dimensions

C. To retrieve the color value of a specific pixel

D. To save the image

Answer: C

Justification: getpixel() returns the color value stored at the specified pixel location.

Q2. What is the purpose of putpixel()?

A. To create a new image

B. To modify the color of a specific pixel

C. To compress the image

D. To display the image

Answer: B

Justification: putpixel() updates the color value of a specified pixel.

Q3. What is the purpose of Image.new()?

A. To open an existing image

B. To copy an existing image

C. To convert an image into grayscale

D. To create a new blank image

Answer: D

Justification: Image.new() creates a new image with the specified mode, size, and background color.

Q4. Why is the save() method used?

A. To store the processed image as a file

B. To display the image on the screen

C. To remove unwanted pixels

D. To count the number of pixels

Answer: A

Justification: The save() method writes the image object to a file.

Q5. (MSQ) Which of the following statements are correct?

A. getpixel() changes the color of a pixel.

B. Image.new() creates a new image with the specified mode and dimensions.

C. putpixel() automatically saves the modified image to disk.

D. Changes made using putpixel() are stored in memory until the image is saved.

Answer: B and D

Justification: Image.new() creates a new image object, and modifications made using putpixel() become permanent only after calling save().

Week 10 – Assignment

Scenario - FLAMES

A school is conducting a fun activity where students enter two names to predict their relationship using the classic FLAMES game. The program removes all common letters from the two names, counts the remaining letters, and repeatedly eliminates letters from the word FLAMES until only one letter remains. Based on the final letter, the corresponding relationship is displayed.

Study the program below and answer the questions.

```
1. name1 = input("Enter first name: ").lower().replace(" ", "")
2. name2 = input("Enter second name: ").lower().replace(" ", "")

3. list1 = list(name1)
4. list2 = list(name2)

5. for ch in name1:
6.     if ch in list2:
7.         list1.remove(ch)
8.         list2.remove(ch)

9. count = len(list1) + len(list2)

10. flames = ["F", "L", "A", "M", "E", "S"]

11. while len(flames) > 1:
12.     index = (count - 1) % len(flames)
13.     flames.pop(index)
14.     flames = flames[index:] + flames[:index]

15. result = {
16.     "F": "Friends",
17.     "L": "Love",
18.     "A": "Affection",
19.     "M": "Marriage",
20.     "E": "Enemies",
21.     "S": "Siblings"
22. }

23. print(result[flames[0]])
```

Question 1 (MCQ)

What is the purpose of the statements in Lines 1 and 2?

A. Convert both names to uppercase.

B. Convert names to lowercase and remove spaces.

C. Reverse both names.

D. Remove duplicate letters from the names.

Answer: B

Justification: The methods lower() and replace(" ", "") convert names to lowercase and remove spaces.

Question 2 (MCQ)

What is stored in the variable count at Line 9?

A. Total number of letters in the original names.

B. Number of common letters.

C. Total number of remaining letters after removing common letters.

D. Number of letters in the word FLAMES.

Answer: C

Justification: count is calculated using the lengths of the remaining characters in both lists.

Question 3 (MCQ)

Which statement actually removes a letter from the FLAMES list during each iteration?

A. Line 11

B. Line 12

C. Line 13

D. Line 14

Answer: C

Justification: pop(index) removes the element at the specified position.

Question 4 (MCQ)

Why is the modulo operator (%) used in Line 12?

A. To convert the index into a floating-point value.

B. To ensure the index always stays within the current size of the list.

C. To sort the FLAMES list.

D. To count the number of common letters.

Answer: B

Justification: The modulo operation wraps the index so that it always remains valid for the current list size.

Question 5 (MSQ)

Which of the following statements are correct about the program?

A. The loop in Lines 11–14 continues until only one letter remains in the flames list.

B. The dictionary result maps each FLAMES letter to a relationship.

C. The program removes all vowels before calculating the result.

D. list(name1) in Line 3 converts the first name into a list of characters.

Answers: A, B, D

Justification:

- A: The loop stops only when one element remains.
- B: The dictionary stores the relationship corresponding to each FLAMES letter.
- D: list(name1) creates a list containing individual characters of the first name.

Scenario – FLAMES 2

A software company has developed a Team Compatibility Analyzer to compare the names of two employees working on a project. The program removes the common letters from both names, counts the remaining letters, and repeatedly eliminates options from a predefined list until only one category remains. The final category is then displayed as the team's compatibility result.

Study the program carefully and answer the following questions.

```
1. name1 = input("Enter first name: ").lower().replace(" ", "")
2. name2 = input("Enter second name: ").lower().replace(" ", "")

3. list1 = list(name1)
4. list2 = list(name2)

5. for ch in name1:
6.     if ch in list2:
7.         list1.remove(ch)
8.         list2.remove(ch)

9. count = len(list1) + len(list2)

10. flames = ["F", "L", "A", "M", "E", "S"]

11. while len(flames) > 1:
12.     index = (count - 1) % len(flames)
13.     flames.pop(index)
14.     flames = flames[index:] + flames[:index]

15. result = {
16.     "F": "Friendly",
17.     "L": "Leader-Follower",
18.     "A": "Adaptive",
19.     "M": "Mentor-Mentee",
20.     "E": "Efficient",
21.     "S": "Supportive"
22. }

23. print(result[flames[0]])
```

Question 1 (MCQ)

Why are the methods lower() and replace(" ", "") used in Lines 1 and 2 before comparing the names?

A. To sort the letters in each name.

B. To make the comparison independent of letter case and spaces.

C. To remove duplicate characters.

D. To reverse the names before processing.

Answer: B

Justification: Converting to lowercase and removing spaces ensures consistent comparison.

Question 2 (MCQ)

Which statement correctly describes the value stored in count at Line 9?

A. The total number of characters entered by the user.

B. The number of matching characters between the two names.

C. The total number of characters remaining after common letters are removed.

D. The number of elements in the flames list.

Answer: C

Justification: count is calculated using the lengths of the updated character lists.

Question 3 (MCQ)

Which line changes the size of the flames list during each iteration of the loop?

A. Line 11

B. Line 12

C. Line 13

D. Line 14

Answer: C

Justification: pop(index) removes one element, reducing the list size by one.

Question 4 (MCQ)

Suppose count is greater than the current length of the flames list. Why does the program still work correctly?

A. Python automatically increases the size of the list.

B. The modulo operator (%) converts the value into a valid list index.

C. The list is sorted before accessing an element.

D. The while loop ignores values larger than the list size.

Answer: B

Justification: The modulo operator wraps the value within the valid index range.

Question 5 (MSQ)

Which of the following statements about the program are correct?

A. The while loop continues until only one element remains in the flames list.

B. Removing Line 14 would change the sequence in which elements are eliminated.

C. The dictionary result is used to remove elements from the flames list.

D. The value of count is computed only after common letters have been removed from both names.

Answers: A, B, D

Justification:

- A: The loop terminates only when one element remains.
- B: Line 14 rotates the list, affecting the elimination order.

- D: count is calculated after updating both character lists.

Scenario: Wild Life

A wildlife research team is studying drone images of a forest to estimate the amount of sunlight reaching the ground. They use a Python program that examines every pixel in an image. For each pixel, the program calculates the average intensity of its Red (R), Green (G), and Blue (B) values. If the average intensity is greater than 127, the pixel is classified as Bright; otherwise, it is classified as Dark. Finally, the program displays the total number of bright and dark pixels in the image.

Study the following program and answer the questions.

```
1. from PIL import Image

2. img = Image.open("sample.jpg")

3. pixels = img.load()

4. width, height = img.size

5. bright = 0
6. dark = 0

7. for x in range(width):
8.     for y in range(height):
9.         r, g, b = pixels[x, y]

10.         average = (r + g + b) // 3

11.         if average > 127:
12.             bright += 1
13.         else:
14.             dark += 1

15. print("Bright Pixels:", bright)
16. print("Dark Pixels:", dark)
```

Question 1 (MCQ)

What is the purpose of Line 3 (pixels = img.load())?

A. It opens the image file.

B. It creates a pixel access object to read individual pixel values.

C. It converts the image into grayscale.

D. It returns the width and height of the image.

Answer: B

Justification: img.load() provides access to the RGB values of individual pixels.

Question 2 (MCQ)

What is stored in the variables width and height at Line 4?

A. The total number of pixels in the image.

B. The RGB values of the first pixel.

C. The dimensions of the image in pixels.

D. The brightness and darkness counts.

Answer: C

Justification: img.size returns the width and height of the image.

Question 3 (MCQ)

What is the value of r, g, and b after executing Line 9?

A. The row, column, and pixel number.

B. The Red, Green, and Blue intensity values of the current pixel.

C. The width, height, and brightness of the image.

D. Three randomly generated numbers.

Answer: B

Justification: pixels[x, y] returns the RGB values of the pixel at position (x, y).

Question 4 (MCQ)

Why is integer division (//) used in Line 10?

A. To compute the average RGB intensity as a whole number.

B. To increase the brightness of every pixel.

C. To convert the image into grayscale.

D. To avoid using nested loops.

Answer: A

Justification: // computes the average as an integer by discarding the decimal part.

Question 5 (MCQ)

Suppose an image has 200 × 100 pixels. How many times will the statement in Line 9 execute?

A. 200

B. 100

C. 300

D. 20,000

Answer: D

Justification: The nested loops visit every pixel once, so the statement executes 200 × 100 = 20,000 times.

Scenario: PixelMap Image Quality Inspector

A photography studio has developed a PixelMap Image Quality Inspector to identify overexposed regions in digital photographs. The program scans every pixel in an image using the PixelMap, calculates the average intensity of its Red (R), Green (G), and Blue (B) values, and classifies each pixel as Bright or Dark based on a threshold value. At the end, it reports the total number of bright and dark pixels.

Study the following program and answer the questions.

```
1. from PIL import Image

2. img = Image.open("sample.jpg")

3. pixels = img.load()

4. width, height = img.size

5. bright = 0
6. dark = 0

7. for x in range(width):
8.     for y in range(height):
9.         r, g, b = pixels[x, y]

10.        average = (r + g + b) // 3

11.        if average > 127:
12.            bright += 1
13.        else:
14.            dark += 1

15. print("Bright Pixels:", bright)
16. print("Dark Pixels:", dark)
```

Question 1 (MCQ)

Which statement ensures that every pixel in the image is processed exactly once?

A. Line 3

B. Lines 7–8

C. Line 10

D. Line 15

Answer: B

Justification: The nested loops iterate through every (x, y) pixel position in the image.

Question 2 (MCQ)

What is the data type of the value returned by pixels[x, y] for a color image?

A. Integer

B. String

C. Tuple

D. List

Answer: C

Justification: A color pixel is returned as an RGB tuple containing three values.

Question 3 (MCQ)

If the current pixel has RGB values (90, 150, 180), what will be the value of average?

A. 120

B. 140

C. 150

D. 420

Answer: B

Justification: (90 + 150 + 180) // 3 = 420 // 3 = 140.

Question 4 (MCQ)

If a pixel has an average intensity of 127, how will it be classified?

A. Bright

B. Dark

C. Both Bright and Dark

D. The program generates an error.

Answer: B

Justification: The condition is average > 127, so 127 is classified as Dark.

Question 5 (MSQ)

Which of the following statements are correct?

A. img.load() allows the program to access individual pixel values.

B. The variables bright and dark are used as counters.

C. The nested loops execute once for each row only.

D. The program processes pixels using their (x, y) coordinates.

Answers: A, B, D

Justification:

- A: img.load() provides pixel-level access.
- B: bright and dark count the number of classified pixels.
- D: Pixels are accessed using their x and y coordinates.

Week 11 – Assignment

Scenario: Semester Planner

A faculty coordinator is preparing the academic schedule for a semester. The coordinator wants to display the calendar of a selected month and identify the weekday on which an important event is scheduled. The following Python program is used.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5. day = int(input("Enter the event date: "))
6.
7. print("Semester Planner")
8. print("-" * 25)
9. print(calendar.month(year, month))
10.
11. weekday = calendar.weekday(year, month, day)
12. days = ["Monday", "Tuesday", "Wednesday", "Thursday",
13.         "Friday", "Saturday", "Sunday"]
14.
15. print("Event Day:", days[weekday])
```

Question 1 (MCQ)

Which function is used to display the calendar of the selected month?

A. calendar.weekday()

B. calendar.month()

C. calendar.date()

D. calendar.day()

Answer: B

Justification: calendar.month() returns the calendar for the specified month and year.

Question 2 (MCQ)

What is stored in the variable weekday after Line 11 executes?

A. The name of the weekday

B. The entered date

C. The index representing the weekday

D. The month number

Answer: C

Justification: calendar.weekday() returns an integer from 0 to 6 representing the weekday.

Question 3 (MCQ)

Why is the list days created in Lines 12–13?

A. To store all months of the year

B. To convert the weekday index into its corresponding weekday name

C. To count the number of weeks in the month

D. To validate the entered date

Answer: B

Justification: The list maps weekday indices to their corresponding weekday names.

Question 4 (MCQ)

Which statement displays the weekday name of the event?

A. print(calendar.month(year, month))

B. weekday = calendar.weekday(year, month, day)

C. print("Semester Planner")

D. print("Event Day:", days[weekday])

Answer: D

Justification: The statement accesses the weekday name using the index and prints it.

Question 5 (MSQ)

Which of the following statements about the program are correct?

A. The user enters the year, month, and day.

B. calendar.weekday() returns the name of the weekday.

C. The list days is used to obtain the weekday name from the returned index.

D. The calendar is displayed before the weekday is printed.

Answer: A, D

Justification: The program accepts three inputs, displays the month's calendar, and then prints the weekday name using the list.

Scenario: Meeting Schedule Planner

An office administrator is preparing the schedule for monthly team meetings. The administrator wants to display the calendar of a selected month, determine whether the year is a leap year, and count the number of Mondays in that month. The following Python program is used.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5.
6. print("Meeting Schedule")
7. print(calendar.month(year, month))
8.
9. if calendar.isleap(year):
10.     print("Leap Year")
11. else:
12.     print("Not a Leap Year")
13.
14. weeks = calendar.monthcalendar(year, month)
15. monday_count = 0
16.
17. for week in weeks:
18.     if week[0] != 0:
19.         monday_count += 1
20.
21. print("Number of Mondays:", monday_count)
```

Question 1 (MCQ)

What is the purpose of calendar.isleap(year)?

A. To display the calendar of the year

B. To check whether the given year is a leap year

C. To count the number of days in the month

D. To find the weekday of the first day of the month

Answer: B

Justification: calendar.isleap() returns True if the given year is a leap year.

Question 2 (MCQ)

What is stored in the variable weeks after Line 14?

A. A list containing the weeks of the month

B. The number of weeks in the month

C. A string representing the calendar

D. The weekday of the first day

Answer: A

Justification: calendar.monthcalendar() returns a list of weeks, where each week is a list of dates.

Question 3 (MCQ)

Why does the program check if week[0] != 0?

A. To count only the weeks with at least one holiday

B. To count only the Mondays that exist in the selected month

C. To skip weekends

D. To find the first week of the month

Answer: B

Justification: A value of 0 indicates that the corresponding Monday does not belong to the selected month.

Question 4 (MCQ)

Which variable stores the total number of Mondays in the selected month?

A. weeks

B. month

C. weekday

D. monday_count

Answer: D

Justification: The variable monday_count is incremented whenever a valid Monday is found.

Question 5 (MSQ)

Which of the following statements are correct?

A. calendar.monthcalendar() returns a list of weeks.

B. The program counts the number of Mondays in the selected month.

C. calendar.isleap() returns the number of days in February.

D. The calendar is displayed before the number of Mondays is printed.

Answer: A, D

Justification: The program first displays the calendar and later prints the Monday count, while monthcalendar() returns a list of weeks.

Scenario: Shift Allocation Calendar

A manufacturing company prepares employee shift schedules every month. Before assigning shifts, the HR team wants to display the calendar, determine the weekday of the first working

day of the month, and count how many weekends (Saturdays and Sundays) occur during that month. The following Python program is used.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5.
6. print("Shift Allocation Calendar")
7. print(calendar.month(year, month))
8.
9. weeks = calendar.monthcalendar(year, month)
10. first_day = calendar.weekday(year, month, 1)
11.
12. day_names = ["Monday", "Tuesday", "Wednesday",
13.              "Thursday", "Friday", "Saturday", "Sunday"]
14.
15. weekend_count = 0
16.
17. for week in weeks:
18.     if week[5] != 0:
19.         weekend_count += 1
20.     if week[6] != 0:
21.         weekend_count += 1
22.
23. print("First day of the month:", day_names[first_day])
24. print("Total weekends:", weekend_count)
```

Question 1 (MCQ)

What is the value stored in first_day after Line 10 executes?

A. The name of the first weekday of the month

B. The date of the first Monday in the month

C. An integer representing the weekday of the first day of the month

D. The total number of weekdays in the month

Answer: C

Justification: calendar.weekday() returns an integer from 0 to 6 representing the weekday.

Question 2 (MCQ)

Why are the conditions week[5] != 0 and week[6] != 0 checked inside the loop?

A. To count Saturdays and Sundays that belong to the selected month

B. To count all working days in the month

C. To find the first weekend of the month

D. To identify public holidays

Answer: A

Justification: A value of 0 indicates that the corresponding date lies outside the selected month.

Question 3 (MCQ)

Which statement best describes the variable weeks?

A. It stores the names of all weekdays.

B. It stores a two-dimensional list where each inner list represents one week.

C. It stores the total number of weeks in the month.

D. It stores all weekend dates only.

Answer: B

Justification: calendar.monthcalendar() returns a list of week lists.

Question 4 (MCQ)

If a selected month contains 5 Saturdays and 4 Sundays, what value will be printed for weekend_count?

A. 4

B. 5

C. 8

D. 9

Answer: D

Justification: The program counts every Saturday and every Sunday separately, resulting in 5 + 4 = 9.

Question 5 (MSQ)

Which of the following statements are correct?

A. calendar.weekday(year, month, 1) is used to determine the weekday of the first day of the month.

B. Each element of weeks is a string representing one week.

C. The program displays the calendar before counting the weekends.

D. The value of weekend_count can never be greater than 8.

Answer: A, C

Justification: The program prints the calendar before iterating through the weeks, and calendar.weekday() returns the weekday index of the specified date.

Scenario: Training Session Scheduler

A training coordinator is planning employee training sessions for a selected month. The program displays the calendar, identifies all Fridays in the month, and determines the last Friday available for conducting a monthly review meeting.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5.
6. print("Training Session Scheduler")
7. print(calendar.month(year, month))
8.
9. weeks = calendar.monthcalendar(year, month)
10. fridays = []
11.
12. for week in weeks:
13.     if week[4] != 0:
14.         fridays.append(week[4])
15.
16. print("Fridays:", fridays)
17.
18. if len(fridays) >= 4:
19.     print("Last Friday:", fridays[-1])
20. else:
21.     print("Insufficient Fridays")
```

Question 1 (MCQ)

What is stored in the list fridays after the loop completes?

A. The weekday numbers of all Fridays

B. The dates that fall on Fridays in the selected month

C. The week numbers containing Fridays

D. The total number of Fridays

Answer: B

Justification: Each valid Friday date is appended to the list.

Question 2 (MCQ)

Why is the condition week[4] != 0 used?

A. To ignore holidays that fall on Fridays

B. To ensure the Friday belongs to the selected month

C. To identify leap years

D. To count only the first Friday

Answer: B

Justification: A value of 0 represents a day outside the selected month.

Question 3 (MCQ)

What is the purpose of fridays[-1] in Line 19?

A. It returns the first Friday of the month.

B. It returns the total number of Fridays.

C. It returns the last element in the fridays list.

D. It removes the last Friday from the list.

Answer: C

Justification: Negative index -1 accesses the last element of a list.

Question 4 (MCQ)

Suppose the selected month has Friday dates 3, 10, 17, 24, and 31. What will be printed by Line 19?

A. 24

B. 17

C. 5

D. 31

Answer: D

Justification: The last element of the list [3, 10, 17, 24, 31] is 31.

Question 5 (MSQ)

Which of the following statements are correct?

A. calendar.monthcalendar() returns a string containing the monthly calendar.

B. week[4] represents the fifth week of the month.

C. The list fridays stores the dates that fall on Fridays.

D. The program displays the last Friday only if at least four Fridays are present in the month.

Answer: C, D

Justification: The list stores Friday dates, and the last Friday is printed only when the list contains at least four elements.

Week 12 – Assignments

Scenario: Ranking the Most Influential Campus Clubs

A university has created a student portal where every club has its own webpage. Each club page contains links to other club pages that it collaborates with or recommends. The university wants the homepage to display the most influential clubs first, not simply those with the highest number of incoming links. They decide to use the PageRank algorithm because it considers both the quantity and quality of incoming links. Clubs that receive recommendations from already influential clubs are expected to receive higher rankings. Answer the following questions based on this scenario.

Q1. Why is the university using PageRank instead of simply counting the number of incoming links?

A. Because it considers the importance of the pages providing the links.

B. Because it ranks pages alphabetically.

C. Because it only counts outgoing links.

D. Because it ignores linked pages completely.

Answer: A

Justification: PageRank assigns greater value to links coming from highly ranked pages.

Q2. Suppose a newly created club page receives a recommendation from a highly ranked club. What is the most likely outcome?

A. The new club will always become the highest-ranked page immediately.

B. The recommendation has no effect because the page is new.

C. The new club's PageRank is likely to improve due to the influential incoming link.

D. The recommending club loses all of its PageRank.

Answer: C

Justification: Links from influential pages contribute positively to the recipient's PageRank.

Q3. If one club page links to hundreds of other club pages, how does this affect the value of each outgoing link?

A. Each outgoing link becomes more valuable.

B. Each outgoing link shares only a smaller portion of the page's ranking value.

C. Outgoing links are ignored by PageRank.

D. The linked pages receive identical final rankings.

Answer: B

Justification: A page distributes its ranking value across all of its outgoing links.

Q4. Which statement best describes how PageRank determines page importance?

A. Pages with the longest content always rank highest.

B. Pages created first always receive the highest rank.

C. Pages with the largest file size receive better rankings.

D. Importance is determined iteratively based on the network of incoming links.

Answer: D

Justification: PageRank repeatedly updates scores until the rankings stabilize.

Q5. (MSQ) Which of the following statements about PageRank are correct?

A. A link from an influential page generally contributes more than a link from an unimportant page.

B. The algorithm depends only on the total number of incoming links.

C. The ranking values are refined over multiple iterations.

D. Every outgoing link from a page transfers an equal share of that page's ranking value.

Answer: A, C

Justification: PageRank values influential links and computes rankings iteratively until convergence.

Scenario: Ranking Webpages Using a Simplified PageRank Score

A search engine developer is testing a simplified version of the PageRank algorithm. Each webpage starts with an initial score of 1.0. During one update cycle, every page distributes its score equally among all the pages it links to. The following program computes the contribution received by each page after one iteration.

```
1. pages = ['A', 'B', 'C', 'D']
2. links = {
3.     'A': ['B', 'C'],
4.     'B': ['C'],
5.     'C': ['A', 'B', 'D'],
6.     'D': ['A']
7. }
8.
9. rank = {'A':1.0, 'B':1.0, 'C':1.0, 'D':1.0}
10. new_rank = {'A':0, 'B':0, 'C':0, 'D':0}
11.
12. for page in pages:
13.     share = rank[page] / len(links[page])
14.     for dest in links[page]:
15.         new_rank[dest] += share
16.
17. print(new_rank)
```

Answer the following questions based on the program.

Q1. What is the purpose of line 13?

A. It counts the number of webpages.

B. It divides the page's current score equally among its outgoing links.

C. It assigns a random score to each page.

D. It resets the PageRank values.

Answer: B

Justification: Each page contributes an equal share of its score to every linked page.

Q2. Which page receives contributions from the greatest number of different pages?

A. A

B. B

C. C

D. D

Answer: C

Justification: Page C receives contributions from both A and B.

Q3. Which page contributes the smallest amount to each linked page?

A. A

B. B

C. C

D. D

Answer: C

Justification: Page C has three outgoing links, so its score is divided into the smallest equal share.

Q4. Which dictionary stores the updated PageRank values after one iteration?

A. pages

B. links

C. rank

D. new_rank

Answer: D

Justification: new_rank accumulates the contributions computed during the iteration.

Q5. (MSQ) Which of the following statements are correct about the program?

A. A page with more outgoing links distributes a smaller share to each linked page.

B. Every page starts with the same initial PageRank value.

C. The program updates rank directly inside the loop.

D. The program performs exactly one PageRank iteration.

Answer: B, D

Justification: All pages start with rank 1.0, and the program computes only one update cycle using new_rank.

Scenario: Evaluating One Iteration of a Simplified PageRank Algorithm

A search engine engineer is analyzing how PageRank values change after one iteration. Every webpage starts with an equal rank of 1.0. During each iteration, a page distributes its rank equally among all the pages it links to. The engineer uses the following program to calculate the updated ranks after a single iteration.

```
pages = ['P', 'Q', 'R', 'S']

links = {
    'P': ['Q', 'R'],
    'Q': ['P', 'R', 'S'],
    'R': ['S'],
    'S': ['P', 'Q']
}

rank = {page: 1.0 for page in pages}
updated = {page: 0 for page in pages}

for page in pages:
    contribution = rank[page] / len(links[page])
    for neighbour in links[page]:
        updated[neighbour] += contribution

print(updated)
```

Answer the following questions based on the program.

Q1. Which page contributes the largest value to each page it links to?

A. P

B. Q

C. R

D. S

Answer: C

Justification: R has only one outgoing link, so its entire rank is transferred to S.

Q2. Which page receives contributions from the highest number of different pages?

A. P

B. Q

C. R

D. S

Answer: B

Justification: Q receives contributions from both P and S.

Q3. Which page will have the highest updated rank after this iteration?

A. P

B. Q

C. R

D. S

Answer: D

Justification: S receives contributions from Q and the full rank of R, giving it the highest total.

Q4. Why is a separate dictionary (updated) used instead of modifying rank directly inside the loop?

A. To reduce memory usage.

B. To ensure every contribution is computed using the original ranks.

C. To avoid sorting the pages.

D. To prevent duplicate page names.

Answer: B

Justification: Updating ranks in-place would affect later calculations within the same iteration.

Q5. (MSQ) Which of the following statements are correct?

A. A page with fewer outgoing links transfers a larger share to each linked page.

B. The rank dictionary is modified during the iteration.

C. The total contribution distributed by each page is equal to its current rank.

D. All updated ranks are computed before replacing the original ranks.

Answer: A, C, D

Justification: Each page distributes all of its current rank, larger shares occur with fewer outgoing links, and updates are accumulated separately before replacing the original values.

Exploring the Collatz Conjecture

A mathematics club is studying the Collatz Conjecture, which states that for any positive integer:

- If the number is even, divide it by 2.
- If the number is odd, multiply it by 3 and add 1.

The process is repeated until the value becomes 1. A student writes the following Python program to generate the sequence for a given starting number.

Answer the following questions based on the program.

```
1. n = int(input())
2. steps = 0
3.
4. while n != 1:
5.     print(n, end=" ")
6.     if n % 2 == 0:
7.         n = n // 2
8.     else:
9.         n = 3 * n + 1
10.    steps += 1
11.
12. print(1)
13. print(steps)
```

Q1. If the input is 8, how many times is the statement on line 10 executed?

A. 3

B. 4

C. 5

D. 6

Answer: A

Justification: The sequence is 8 → 4 → 2 → 1, requiring 3 iterations.

Q2. Which statement correctly describes the purpose of line 6?

A. It checks whether the current number is prime.

B. It checks whether the current number is even.

C. It checks whether the sequence has ended.

D. It checks whether the number is greater than 1.

Answer: B

Justification: The modulo operation determines whether the current value is even.

Q3. If the input is 3, which number is printed immediately after 10?

A. 8

B. 16

C. 5

D. 20

Answer: C

Justification: The sequence is 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1.

Q4. What is the role of the variable steps?

A. It stores the largest value in the sequence.

B. It counts the number of even numbers encountered.

C. It stores the final value of the sequence.

D. It counts the number of transformations performed before reaching 1.

Answer: D

Justification: steps is incremented once during each iteration of the loop.

Q5. (MSQ) Which of the following statements are correct?

A. The value 1 is printed even though the loop terminates before processing it.

B. For an odd value of n, the next value is always smaller than the current value.

C. The variable steps is incremented exactly once during every loop iteration.

D. The program prints every intermediate value generated by the sequence.

Answer: A, C, D

Justification: The loop prints all intermediate values, 1 is printed after the loop, and steps increases once per iteration.